



Modeling Soft Intervention Effects for Implicit Causal Representation Learning

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Received: 8 August 2025 / Revised: 9 April 2026 / Accepted: 6 August 2026

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Abstract

Learning causal representations from data without access to ground-truth causal graphs remains a central challenge in representation learning. Implicit approaches—where the model learns causal dependencies without explicitly parameterizing the causal graph—offer a compelling advantage over explicit methods by avoiding optimization difficulties such as local minima associated with adjacency matrix estimation. However, existing implicit methods often assume access to hard interventions, which are rarely feasible in real-world scenarios. In contrast, soft interventions—more prevalent in practice—modify causal mechanisms without severing parental dependencies, introducing subtle, ambiguous effects that confound learning. To address this, we propose ICRL-SM, a novel method for Implicit Causal Representation Learning from soft interventions using a causal mechanism switch variable. This variable captures unwanted changes induced by soft interventions, enabling the model to focus on the necessary variations that reflect underlying causal structure. Our framework leverages a variational autoencoder trained on paired pre- and post-intervention samples, and is theoretically grounded under a set of assumptions that ensure identifiability of causal representations. Although some assumptions (e.g., Gaussianity of latent variables, diffeomorphic decoders) are strict, we show empirically that our method performs robustly even when these are violated—demonstrating strong results on both synthetic benchmarks and real-world image datasets. These findings highlight the potential of ICRL-SM to bridge the gap between theoretical identifiability and practical applicability, advancing causal representation learning under realistic conditions. The source code for this paper is available at: <https://github.com/sshirahmad/ICRL>.

Keywords Causal representation learning · Implicit causal model · Soft interventions · Causal mechanisms · Variational autoencoder · Identifiability · Representation learning · Intervention modeling

Editor: Henry Gouk.

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1 Introduction

One of the long-standing challenges in causal representation learning is how to recover the underlying causal variables of a system from high-dimensional observations such as images, videos, or sensor streams. This problem is central to *identifiability* (Kivva et al., 2022): ensuring that the learned latent representations correspond to genuine causal factors rather than arbitrary statistical encodings. Without identifiability, models may achieve strong predictive performance while capturing spurious correlations, leading to brittle behavior under distribution shifts or interventions. The central question is therefore how to learn identifiable causal representations from high-dimensional data when interventions modify underlying mechanisms rather than fully breaking causal links.

This issue is particularly critical in real-world settings where models are expected to generalize across environments—for example, when a robotic system encounters new manipulation conditions, when medical imaging data shifts across acquisition protocols, or when scientific experiments modify underlying mechanisms. In such cases, representations that fail to align with the true causal structure cannot reliably support reasoning under interventions.

This challenge is especially relevant in medicine and drug discovery, where interventions are used to uncover the true causes of biological outcomes. For instance, administering a drug can help determine whether a physiological effect is directly caused by a target factor or influenced by confounding variables. Figure 1 illustrates this principle in a simplified scenario: exercise as the causal factor and heart rate as the effect, with hard and soft interventions (beta-blocker and external factors such as caffeine) showing how interventions can modify observed outcomes and reveal causal relationships. Accurately capturing these

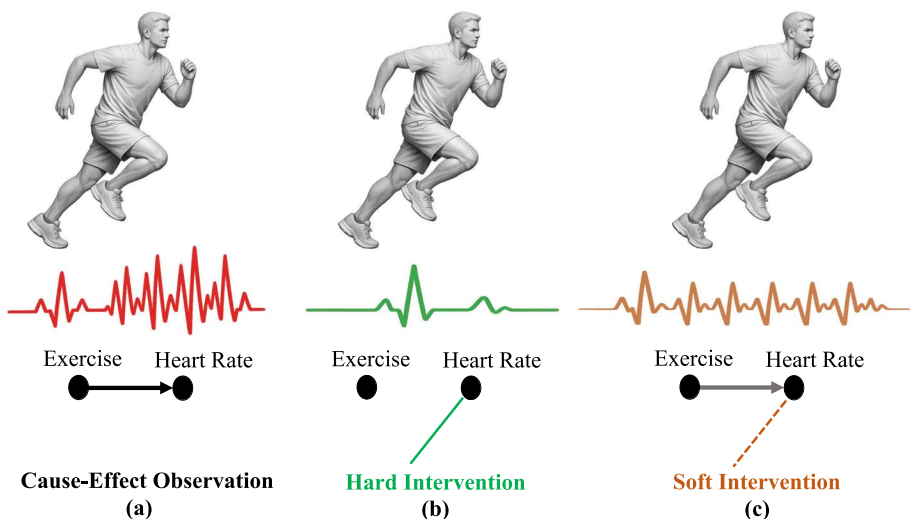


Fig. 1 Illustration of causal interventions on heart rate. **a** Observed: Heart rate naturally increases with exercise in a controlled population. **b** Hard intervention: Beta-blocker suppresses heart rate completely under controlled conditions (no caffeine, chocolate, or other stimulants), breaking the causal effect of exercise. **c** Soft intervention: Beta-blocker is applied in a non-controlled environment where participants may consume caffeine or other stimulants, partially restoring heart rate variation while still reducing the effect of exercise compared to the no-intervention scenario

interventions is essential for models to reason reliably about underlying mechanisms in both biomedical and experimental settings.

A key theoretical barrier is that purely observational data can only identify a Markov Equivalence Class (MEC) of causal graphs, where multiple graph structures induce indistinguishable observational distributions (Schölkopf, 2022). Resolving this ambiguity requires additional assumptions or sources of supervision, such as access to ground-truth causal variables (Yang et al., 2021), known model parameters (Ahuja et al., 2023), paired interventional data (Brehmer et al., 2022; Kügelgen et al., 2021), or explicit knowledge of intervention targets (Lippe et al., 2022). However, many real-world interventions are *soft*: they modify structural mechanisms without severing causal dependencies, and their precise effects are often only partially observed.

In this paper, we introduce ICRL-SM (Implicit Causal Representation Learning via Switchable Mechanisms), a novel framework designed to address this gap by explicitly modeling soft interventions at the level of structural mechanisms. Our key innovation is the introduction of a causal mechanism switch variable, which explicitly models the noise induced by soft interventions. This variable enables the model to isolate intervention-relevant variation from noise effects, leading to more accurate recovery of underlying causal representations.

Our method is grounded in a VAE-based architecture and learns from paired pre- and post-intervention samples. We provide assumptions under which causal representations can be identified up to reparameterization. While some of these assumptions may not strictly hold in all real-world datasets, we demonstrate empirically that ICRL-SM remains effective and robust even when they are partially violated.

We evaluate our approach on both synthetic datasets and real-world image domains. Experimental results show that ICRL-SM consistently outperforms baselines in recovering meaningful causal representations. These results underscore the potential of implicit causal learning methods for broader applicability in complex, intervention-rich environments.

Our key contributions can be summarized as follows:

1. We propose ICRL-SM, a new method for implicit causal representation learning that works directly with soft interventions.
2. We introduce the causal mechanism switch variable to explicitly model subtle and unwanted shifts induced by soft interventions.
3. We provide theoretical results showing identifiability up to reparameterization under a set of assumptions.
4. We demonstrate strong empirical performance across both synthetic and real-world image datasets, even when identifiability assumptions are only approximately met.

2 Background

2.1 Interventions

In causal modeling, an *intervention* refers to a deliberate manipulation of one or more variables to assess their causal influence on others within a system. By observing the outcomes of such manipulations, one can infer directional, cause-effect relationships that go beyond

statistical correlation. Interventions are typically categorized based on the level of control they exert over the causal system. Two fundamental types are *Hard* and *Soft* interventions.

Definition 1 Structural Causal Models A structural causal model (SCM) is a tuple $\mathcal{C} = (\mathcal{F}, \mathcal{Z}, \mathcal{E}, \mathcal{G})$ with the following components:

1. The domain of causal variables $\mathcal{Z} = \mathcal{Z}_1 \times \mathcal{Z}_2 \times \dots \times \mathcal{Z}_n$.
2. The domain of exogenous variables $\mathcal{E} = \mathcal{E}_1 \times \mathcal{E}_2 \times \dots \times \mathcal{E}_n$.
3. A directed acyclic graph $\mathcal{G}(\mathcal{C})$ over the causal and exogenous variables.
4. A causal mechanism $f_i \in \mathcal{F}$ which maps an assignment of parent values for the parents Z_{pa_i} plus an exogenous variable value for E_i to a value of causal variable Z_i .

2.1.1 Hard Intervention

Hard interventions involve forcibly setting a variable to a fixed value, thereby severing its dependence on any causal parents. This operation corresponds to the standard do-operator in Structural Causal Models (SCMs) (Definition 1) (Pearl et al., 2016). While theoretically powerful, hard interventions are often impractical in real-world settings due to ethical, technical, or logistical constraints.

A hard intervention effectively replaces the original causal mechanism of a variable Z with a constant value, expressed as $do(Z = z)$. This substitution completely removes the influence of Z 's parents, as illustrated in the Figure 1b. In graphical terms, it corresponds to deleting all incoming edges to the intervened node, reflecting the elimination of upstream causal dependencies. A canonical example is a Randomized Controlled Trial (RCT), where treatments are randomly assigned to participants, thereby eliminating confounding by design.

Formally, if a variable Z_i is determined in the observational setting by a function $f_i(z_{pa_i}, e_i)$ -where z_{pa_i} are its parents and e_i is an exogenous noise term-then under a hard intervention:

$$\tilde{z}_i = z^* \quad (\text{constant}), \quad \text{independent of } z_{pa_i}.$$

This full isolation of the variable from its causal ancestors provides strong identifiability guarantees, but it typically requires strict experimental control that may not be feasible outside of laboratory or clinical environments.

2.1.2 Soft Intervention

In contrast to hard interventions, soft interventions subtly alter the underlying causal mechanisms without fully overriding them. Rather than replacing a causal variable with a fixed value, soft interventions perturb the functional or probabilistic relationships that govern the variable's behavior. Crucially, they preserve the variable's dependence on its parent variables, maintaining the overall graph structure of the system.

Formally, if a variable z_i is generated by $f_i(z_{pa_i}, e_i)$ in the observational regime, a soft intervention modifies this to:

$$p(z_i | z_{pa_i}) \rightarrow \tilde{p}(z_i | z_{pa_i}) \quad \text{with} \quad \tilde{z}_i = \tilde{f}_i(z_{pa_i}, \tilde{e}_i),$$

as shown in Figure 1c, and discussed in Correa et al. (2020). Graphically, the intervened node retains its incoming edges, but its internal mechanism is changed.

These subtler shifts make soft interventions more reflective of real-world scenarios—such as environmental changes, behavioral nudges, or policy shifts—where full control over variables is impractical or impossible. However, they also introduce significant ambiguity. For instance, in the example of Figure 1, a change in heart rate might stem from a soft intervention or simply from preserved parental influence. This overlap complicates the process of inferring causal structure from observational or interventional data.

The key distinction lies in how the interventions affect the system. Hard interventions sever dependencies, offering clean signals for causal discovery. Soft interventions, by preserving dependencies but modifying mechanisms, impose fewer constraints on the learner. As a result, causal inference under soft interventions becomes significantly more challenging. The model must disentangle the underlying changes from subtle variations in conditional distributions, often with limited or ambiguous evidence.

2.1.3 Real-World Motivation

Despite their difficulty, soft interventions are highly relevant to naturalistic domains, including vision, language, healthcare, and behavioral sciences. Figure 2 illustrates real-world image examples where such soft interventional changes manifest—highlighting the importance of models like ICRL-SM, which are designed to learn robust causal representations under these more realistic and complex conditions.

2.2 Causal Graph Structure

Another challenge stems from uncertainty about the causal graph itself. In some settings, domain knowledge can be used to define the causal structure a priori, enabling the learning of corresponding causal variables (Bagi et al., 2023; Liu et al., 2021, 2022; Li et al., 2025). However, in many practical applications, the graph must be learned jointly with the representations. This has motivated two major lines of work in causal representation learning within the Variational Autoencoder (VAE) framework: Explicit Latent Causal Models (ELCMs) and Implicit Latent Causal Models (ILCMs).

2.2.1 Explicit Latent Causal Models

Explicit Latent Causal Models (ELCMs) learn both the causal graph structure and the latent causal representations by parameterizing the adjacency matrix and enforcing a factorized prior that reflects the assumed Directed Acyclic Graph (DAG) structure (Yang et al., 2021; Ahuja et al., 2023; Lippe et al., 2022; Yang et al., 2021; Yu et al., 2019; Lachapelle et al., 2022; Ni et al., 2025). In these models, each latent variable corresponds to a causal variable, and its distribution is conditioned on its parents in the graph. The overall prior over latent variables is thus factorized as:

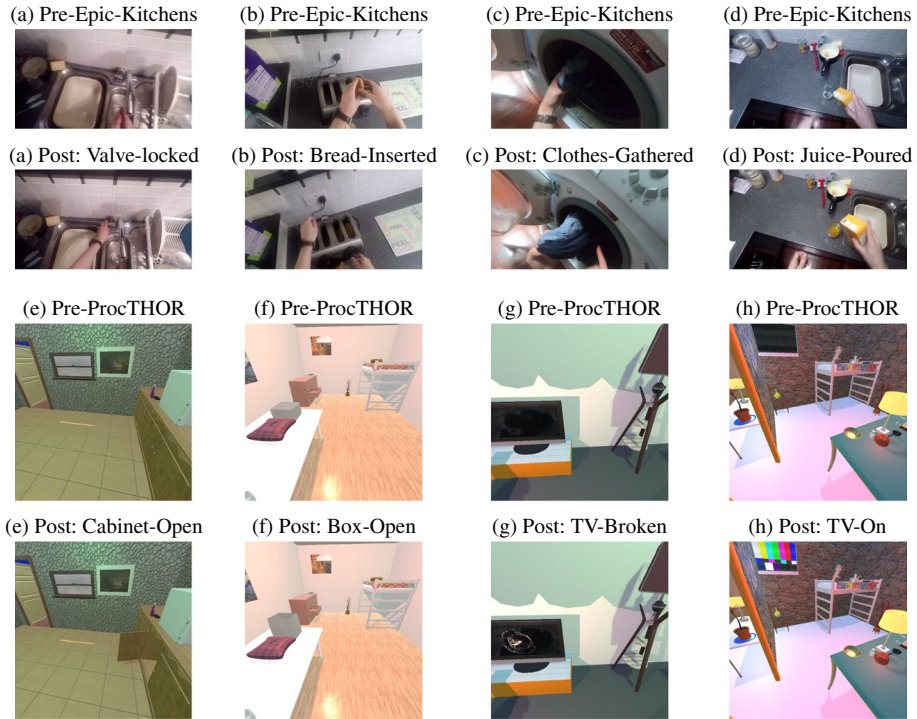


Fig. 2 In the Causal-Triplet dataset (Liu et al., 2023), visual representations capture both pre and post-intervention scenarios. The first two rows showcase data samples from Epic-Kitchens, while the third and fourth rows feature samples from ProcTHOR. Each image in the post-intervention condition is accompanied by labels specifying the corresponding action and intervened object. In the images in the first two rows, the agent is performing an action on an object but the camera angle has also changed. So we can say that for example the distribution of causal variables conditioned on the camera angle has been changed due to soft intervention

$$p(z) = \prod_i p(z_i \mid Z_{pa_i}). \quad (1)$$

This explicit factorization encourages interpretability, as the learned structure directly reflects hypothesized causal relationships. Parent-child dependencies are encoded into the prior, enabling targeted interventions and causal reasoning.

However, ELCMs often face significant optimization challenges. The learning of the graph structure and the latent representations is tightly coupled which can lead to local minima and unstable training dynamics. Small changes in one component (e.g., adjacency structure) can drastically affect the other (e.g., latent encoding), making convergence difficult.

2.2.2 Implicit Latent Causal Models

Before proceeding further, we introduce a simple SCM to build intuition behind the use of ILCMs:

$$\begin{cases} z_1 = f_1(e_1) \\ z_2 = f_2(z_1, e_2) \\ z_3 = f_3(z_2, e_3) \end{cases} \quad (2)$$

This SCM can be viewed as a system of structural equations, where each causal variable z_i is determined by its parents and an independent exogenous noise terms e_i . By recursively substituting upstream variables, we can express each z_i directly in terms of the exogenous variables:

$$\begin{cases} z_1 = s_1(e_1) \\ z_2 = s_2(e_2; e_1) \\ z_3 = s_3(e_3; e_1, e_2) \end{cases} \quad (3)$$

Equation 3 shows that each causal variable can be written as a solution function s_i of its ancestral exogenous variables (Brehmer et al., 2022). The specific parameterization of these solution functions is introduced later in Section 4.6.

Importantly, ILCMs do not require learning the adjacency matrix explicitly. Instead, the model includes all exogenous variables as inputs to the solution functions and learns the relevant dependencies during training. This approach avoids the entanglement between graph structure learning and representation learning that often hinders optimization in ELCMs.

In contrast to ELCMs, which jointly learn both the graph and latent structure, ILCMs side-step direct graph estimation by modeling causal dependencies implicitly through the solution functions. This leads to more stable and scalable training, albeit at the cost of reduced structural interpretability (Brehmer et al., 2022). However, the existing ILCMs have largely focused on hard interventions, limiting their relevance in settings where only soft interventions are available. Soft interventions complicate the learning process by introducing subtle, non-deterministic shifts that are difficult to distinguish from background variability.

3 Related Work

Causal representation learning has recently garnered significant attention (Schölkopf et al., 2021; Kaddour et al., 2022). The primary challenge in this problem lies in achieving identifiability beyond the Markov equivalence class (Schölkopf, 2022). Solely relying on observational data necessitates additional assumptions regarding causal mechanisms, decoders, latent structure, and the availability of interventional data (Yang et al., 2021; Ahuja et al., 2023; Lachapelle et al., 2022; Lu et al., 2022; Shen et al., 2022; Yu et al., 2020; Perry et al., 2022; Zheng et al., 2018; Jaber et al., 2020). Recent works have focused on identifying causal models from collected interventional data instead of making strong assumptions about functions of the causal model. Interventional data facilitates identifiability based on relatively weak assumptions (Ahuja et al., 2023; Brehmer et al., 2022; Cooper & Yoo, 2013; Zhang et al., 2023; Wendong et al., 2023). This type of data can be further categorized based on whether it involves soft or hard interventions, and whether the manipulated variables are observed and specified or latent. Our focus in this paper is on examining soft interventions, encompassing both observed and unobserved variables.

3.1 Explicit Models vs. Implicit Models

Table 1 presents a comparison of the assumptions and identifiability results between our proposed theory and other related works on causal representation learning with interventions.

In causal representation learning with interventions, one approach assumes a given causal graph and concentrates on identifying causal mechanisms and mixing functions. For instance, Causal Component Analysis (CauCA) (Wendong et al., 2023) explores soft interventions with a known graph. Alternatively, when the graph is not provided, explicit models seek to reconstruct it from interventional data (Lippe et al., 2022; Cooper & Yoo, 2013), potentially resulting in a chicken-and-egg problem in causal representation learning (Brehmer et al., 2022).

Current methods face the challenge of simultaneously learning the causal graph and other network parameters, especially in the absence of information about causal variables or the graph. Addressing these challenges, Brehmer et al. (2022) recently introduced ILCM, which performs *implicit* causal representation learning exclusively using *hard* intervention data. In contrast, our approach introduces a novel method for learning an implicit model from *soft* interventions. Brehmer et al. (2022) describes methods for extracting a causal graph from a learned implicit model, which could be applied to our method as well. In our experiments, we will compare our method with ILCM and disentangled VAE (dVAE (Locatello et al., 2020)), given their implicit nature and similar experimental settings and assumptions. Additionally, to showcase the superiority of our method over explicit models, we will employ explicit causal model discovery methods like ENCO (Lippe et al., 2022) and DDS (Charpentier et al., 2022), in conjunction with various variants of β -VAE (Higgins et al., 2017).

Table 1 Comparison of proposed method with other recent related work on causal learning from interventional data

Methods	Causal Mechanism	Mixing Function	Intervention Type	Learning Method	Identifiability guarantees
Causal-Disc (Zhang et al., 2023)	Nonlinear	Full row rank polynomial	Soft	Explicit	Permutation & Affine
CauCA (Wendong et al., 2023)	Nonlinear	Diffeomorphism	Soft	Explicit	Different based on assumptions
Linear-CD (Squires et al., 2023)	Linear	Linear	Hard	Explicit	Permutation
LScale-I (Varici et al., 2025)	Linear/Nonlinear	Linear	Hard/Soft	Explicit	Different based on assumptions
ILCM (Brehmer et al., 2022)	Nonlinear	Diffeomorphism	Hard	Implicit	Permutation & reparameterization
dVAE (Locatello et al., 2020)	Nonlinear	Diffeomorphism	Hard	Implicit	Permutation & reparameterization
IRSS (Li et al., 2025) ^a	Nonlinear	Neural network	None	Explicit	N/A
ICRL-SM (ours)	Nonlinear	Diffeomorphism	Soft	Implicit	Reparameterization

^a IRSS addresses distribution shift using interventional data gathered from multiple domains assuming causal variables are independent and are sampled from a Gaussian distribution

3.2 Hard Interventions vs Soft Interventions

The identification of explicit causal models from hard interventions has been extensively explored. Squires et al. (2023) investigates causal disentanglement in linear causal models with linear mixing functions under hard interventions. Similarly, Buchholz et al. (2023) focus on identifying causal models with linear causal mechanisms and nonlinear mixing functions, also utilizing hard interventions. In a more general setting with non-parametric causal mechanisms and mixing functions, Kügelgen et al. (2023) examines the identifiability of causal models, utilizing multi-environment data from unknown interventions. Similarly, Bagi et al. (2023) explores identifiability of causal models using multi-environment data from unknown interventions. Varici et al. (2023) investigates the identifiability of causal models with nonlinear causal mechanisms and linear mixing functions, considering both hard and soft interventions. Varici et al. (2025) investigates the identifiability of causal models with general causal mechanisms and linear mixing functions, considering both hard and soft interventions.

Recent work has expanded the concept of explicit hard interventions to include soft interventions. In their study, Zhang et al. (2023) addresses the identification of causal models from soft interventions, leveraging the sparsity of the adjacency matrix as an inductive bias. However, when dealing with implicit models, soft interventions introduce new complexities. Identifiability becomes more challenging, as the causal effect of variables on observed variables is less apparent. This ambiguity arises from the dual possibility of effects originating from interventions or influences from parent variables on the causal variables. Moreover, in scenarios where implicit modeling is retained, the absence of knowledge about parent variables further complicates identifiability. While Brehmer et al. (2022) theoretically establishes identifiability for hard interventions, practical experiments involving complex causal models with over 10 variables reveal increased ambiguity and confounding factors. Consequently, model identification becomes less straightforward.

4 Methodology

In this section, we introduce our method, ICRL-SM (see Figure 3). Our goal is to recover identifiable causal representations from data influenced by soft interventions, without explicitly learning a causal graph. We begin by describing the data-generating process and key components of our model, followed by a formal statement of assumptions, an explanation of the causal mechanism switch variable, identifiability conditions, training objectives, and implementation of solution functions.

4.1 Notations

For better and easier understanding of our proposed methodology, including the underlying theory and its proof, we adhere to the notation conventions presented in Table 2 throughout the paper.

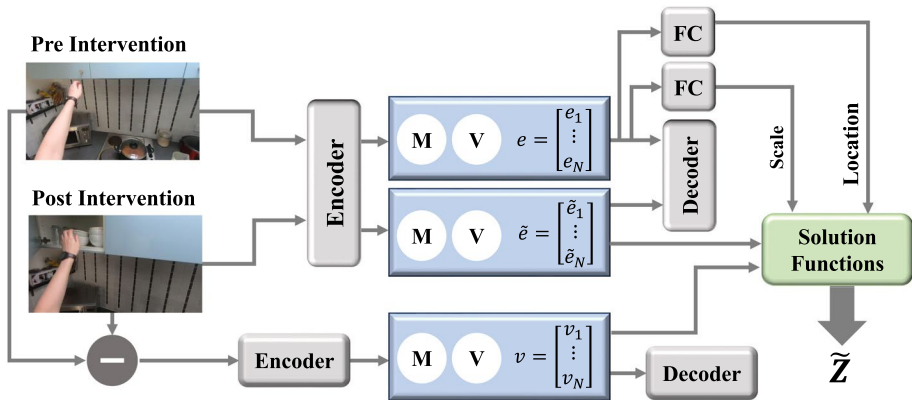


Fig. 3 General overview of ICRL-SM. The architecture processes pre-intervention observations X , post-intervention observations $\tilde{X}$, and their differences (intervention displacement) $\tilde{X} - X$, encoding them into latent representations. Each encoder outputs the mean (M) and variance (V) of a probability distribution. By sampling from these distributions, we derive pre-intervention exogenous variables E , post-intervention exogenous variables $\tilde{E}$, and the causal mechanism switch variable V , with V obtained from the encoding of the differences. The exogenous variables E pass through two fully connected (FC) layers to predict scale and location parameters. These parameters, along with the $\tilde{E}$ and V , are used in the solution function (Equation 24) to compute post-intervention causal variables $\tilde{Z}$. N denotes the total number of causal variables

Table 2 Notation conventions

Symbol	Description
Lowercase letters (x)	Values of variables
Uppercase letters (X)	Variables themselves
Calligraphic letters ($\mathcal{X}$)	Domains of variables and sets
i	General index for variables
$i \in I$	Index for intervention targets among causal variables
$(\tilde{\cdot})$	Post-intervention state
$(\cdot)_{/i}$	Exclusion of the i -th element

4.2 Data Generation Process

We model the data generation process using a SCM (Definition 1) with unobserved causal variables Z , exogenous variables E , and observed variables X .

The observations are generated via a diffeomorphic (Definition 2) mixing function $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$:

$$g(z) = x, \quad (4)$$

Definition 2 (*Diffeomorphism*) A diffeomorphism between smooth manifolds $\mathcal{M}$ and $\mathcal{N}$ is a bijective map $f : \mathcal{M} \rightarrow \mathcal{N}$, which is smooth and has a smooth inverse. Diffeomorphisms

preserve information as they are invertible transformations without discontinuous changes in their image.

Our objective is to learn a latent space representing Z , using paired pre-intervention and post-intervention samples $(x, \tilde{x})$ and known intervention targets $i \in \mathcal{I}$.

Instead of explicitly parameterizing causal graphs (as in explicit latent causal models), we take an implicit approach, directly learning the solution functions s_i that map exogenous variables E and a causal mechanism switch variable V to causal variables Z :

$$s_i : \mathcal{E}_i \times \mathcal{E}_{/i} \times \mathcal{V} \rightarrow \mathcal{Z}_i \quad (5)$$

These functions allow us to model how interventions modify the underlying mechanisms that generate causal variables. We denote the post-intervention solutions as $\tilde{z}_i = s_i(\tilde{e}_i; e_{/i}, v)$, where v models the soft intervention effects.

4.3 Causal Mechanism Switch Variable

4.3.1 Motivation

Our implicit modeling approach operates without requiring explicit knowledge of the causal graph. All exogenous variables are included in the solution functions, and the structure of the causal graph remains unchanged during training. That is, we assume a full causal graph where the existence of edges is fixed, and the solution functions learn the effective dependencies by adjusting the weights of the edges, including the strength of dependence on exogenous variables.

To illustrate the difference between hard and soft interventions under this modeling framework, consider a simple two-variable causal system with $Z_1 \rightarrow Z_2$. Suppose we model Z_2 using a location-scale noise model (LSNM (Immer et al., 2023)) introduced in Section 4.6, without incorporating a mechanism switch variable V :

$$\tilde{z}_2 = \frac{\tilde{e}_2 - \widetilde{\text{loc}}(e_1)}{\widetilde{\text{scale}}(e_1)}, \quad (6)$$

where $\tilde{e}_2$ is exogenous noise for Z_2 under the soft intervention, and $\widetilde{\text{loc}}(e_1)$ and $\widetilde{\text{scale}}(e_1)$ are learned functions of the noise for Z_1 . The resulting mean of the interventional distribution $p(\tilde{Z}_2 \mid E_1 = e_1)$ is:

$$\text{mean}_{p(\tilde{Z}_2 \mid E_1 = e_1)} = \frac{1}{\widetilde{\text{scale}}(e_1)} \times \mathbb{E}[\tilde{E}_2] - \frac{\widetilde{\text{loc}}(e_1)}{\widetilde{\text{scale}}(e_1)}. \quad (7)$$

4.3.1.1 Hard Intervention Under a hard intervention on Z_2 , we assume that all parental influences are severed, so the distribution of Z_2 becomes independent of E_1 . That is,

$$p(\tilde{Z}_2 \mid E_1 = e_1) = p(\tilde{Z}_2) = \mathcal{N}(0, 1). \quad (8)$$

This means that the solution functions will learn to ignore parental variables (e.g., setting loc and scale to constants), reflecting the absence of causal dependence.

4.3.1.2 Soft Intervention In contrast, a soft intervention on Z_2 preserves the influence of E_1 , leading to:

$$p(\tilde{Z}_2 \mid E_1 = e_1) \neq \mathcal{N}(0, 1). \quad (9)$$

This implies that information from E_1 still affects $\tilde{Z}_2$. If we do not account for this, the solution function will incorrectly route parental effects through $\tilde{e}_2$, thereby violating identifiability up to reparameterization.

To address this, we introduce the causal mechanism switch variable V . Rather than letting the exogenous variable E_2 absorb both noise and soft intervention effects, we model soft changes in the causal mechanism (e.g., due to the intervention) via V . In doing so, we redirect parental effects to propagate through V , preserving identifiability and allowing the model to disentangle endogenous influences from intervention-induced changes.

4.3.2 Definition

We introduce the *causal mechanism switch variable* V , which explicitly encodes intervention-induced changes to the structural mechanism of a causal variable Z_i under soft interventions. Formally, the original structural mechanism is given by

$$Z_i = s_i(E_i; E_{/i}), \quad (10)$$

where s_i denotes the solution function. Under a soft intervention, this mechanism is augmented to

$$Z_i = s_i(E_i; E_{/i}, V), \quad (11)$$

4.3.3 Formalization

This insight motivates our proposed framework: Augmented Implicit Causal Models (AICMs). We formally define this class of models as follows:

Definition 3 (*Augmented Implicit Causal Models*) An *Augmented Implicit Causal Model (AICM)* is defined as $\mathcal{A} = (\mathcal{S}, \mathcal{Z}, \mathcal{E}, \mathcal{V})$, where:

- $\mathcal{S}$ is the set of solution functions,
- $\mathcal{Z}$ is the space of endogenous variables,
- $\mathcal{E}$ is the space of exogenous variables, and
- $\mathcal{V}$ is the space of *causal mechanism switch variables*,

which model the effect of soft interventions on the causal mechanisms via the expansion in Equation 12.

4.3.4 Intuition

To address the challenges of causal representation learning under soft interventions, our proposed model (Figure 4) leverages a causal mechanism switch variable V , which explicitly encodes the modifications to causal mechanisms caused by soft interventions. The key idea is to allow the model to flexibly interpolate between pre- and post-intervention mechanisms—even when the underlying causal graph is unknown—thereby disentangling the effects of soft interventions from the exogenous variables. This enables more accurate recovery of intrinsic causal representations.

To understand the role of V , consider a Taylor series expansion of the solution function s_i with respect to v , where v_0 denotes the pre-intervention state:

$$\begin{aligned} s_i(\tilde{e}_i; e_{/i}, v) &= s_i(\tilde{e}_i; e_{/i}, v_0) + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\frac{\partial^n s_i}{\partial v^n} \Big|_{v=v_0} (v - v_0)^n \right) \\ &= s_i(\tilde{e}_i; e_{/i}, v_0) + R_i, \end{aligned} \quad (12)$$

where, in Equation 12:

- $s_i(\tilde{e}_i; e_{/i}, v_0)$: This term represents the *original* (pre-intervention) solution function with the updated exogenous input $\tilde{e}_i$. It captures the true intrinsic behavior of variable Z_i -isolated from unwanted soft intervention effects-analogous to a hard intervention.
- R_i : This term encapsulates the *intervention-induced changes* due to soft interventions. It models deviations from the original function caused by modifications in V .

Our objective is to retain the first term, which reflects the genuine causal content of the variable, while accounting for and mitigating the influence of the second term, which arises from unwanted soft intervention effects.

4.3.5 Observability

The goal of introducing the causal mechanism switch variable V is to disentangle the effect of soft interventions on a latent variable $\tilde{Z}_i$ into two distinct components; (i) A change in the causal mechanism and its parent variables, and (ii) A change in the exogenous variable E_i .

Consider a dataset consisting of images where a human performs actions on household objects-e.g., *fold a towel*. The underlying causal variables in such scenes correspond to

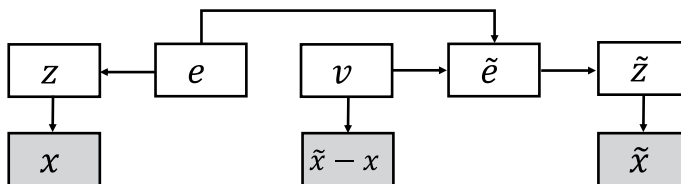


Fig. 4 Generative model. The pre- and post-intervention causal variables generate the pre- and post-intervention observations x and $\tilde{x}$, respectively. The causal variables are obtained from the exogenous variables via the solution functions. The causal mechanism switch variable v models the effects of soft interventions which can be observed from $\tilde{x} - x$ in systems with linear decoder

object attributes such as *shape*, *color*, and *size*. Suppose a soft intervention is applied via the action *fold*, which aims to causally alter the *shape* of the towel. However, the intervention may inadvertently alter other factors, such as the camera angle or lighting. For instance, folding the towel can occlude portions of it or shift the camera's viewpoint slightly, leading to visual differences that are not directly attributable to the action.

These unintentional shifts change the observed pixels in a way that is not mediated solely by the causal path from folding the towel to its shape. If we had access to such nuisance factors (e.g., camera tilt, lighting metadata), we could model these influences explicitly. This motivates the causal mechanism switch variable V : a latent variable encoding how the intervention modifies the system's causal mechanisms beyond its intended target. In this example, V captures nuisance factors such as camera viewpoint or lighting variation, enabling the model to distinguish intrinsic causal changes from extraneous visual shifts. To clarify the role of this variable, we interpret V as follows.

The variable V should be understood as a latent descriptor of how a causal mechanism changes under a soft intervention, rather than a directly measurable physical variable. In the special case of a linear mixing function, the difference between paired observations provides partial information about this mechanism shift, making V approximately observable, as discussed later in this section. However, observability of V obtained from the difference of paired observations is a property of the linear setting and does not generalize to realistic nonlinear generative processes.

In real-world practical scenarios beyond linear mixing functions, suitable data augmentations can be applied. In such cases, V captures the latent transformation describing how the causal mechanism changes under the intervention, even though it cannot be directly computed from pixel differences. Instead, V is inferred implicitly during learning. This distinction highlights that the mathematical derivation for linear mixing function case provides intuition rather than a strict requirement and that additional data augmentations may provide useful signals for learning representations that capture the variability encoded by V . Providing augmentations of observations, we propose a supervisory signal for the model to better learn V .

By observing or inferring V , the model can better estimate the isolated effect of the intervention on the causal variable (e.g., shape) by effectively "switching" between pre- and post-intervention mechanisms-aligning with the structure of Equation 12. For visual intuition, see Figure 2, which contrasts intrinsic changes (due to folding) with extrinsic ones (due to camera angle). Without observability of V , the model must learn a more complex transformation to account for these confounding changes. As a result, the residual term R_i in Equation 12 may become highly nonlinear or dependent on higher-order interactions. Therefore, for tractability, we assume that V can be inferred from the data.

4.3.5.1 Observability Under Linear Decoders Under a linear mixing function, V becomes partially observable via a simple transformation. Given a paired pre/post-intervention triplet $(x, \tilde{x}, i)$ and a linear mixing function g , we can express the soft intervention effect on the latent space as:

$$\tilde{z} - z \stackrel{(12)}{=} \Delta e_i + R, \quad (13)$$

$$\tilde{x} - x \stackrel{(4)}{=} g(\tilde{z}) - g(z) \stackrel{\text{Linearity}}{=} g(\tilde{z} - z) \stackrel{(13)}{=} g(\Delta e_i + R), \quad (14)$$

where $R = [R_0, R_1, \dots, R_n]$ and n is the number of causal variables. Here, R_j denotes the soft intervention effect on the j -th causal mechanism, and Δe_i captures the change in the exogenous variable of the intervened latent factor.

Since interventions are assumed to only affect one causal variable, all components of R except for R_i (the target) are zero. Consequently, when g is linear, the difference $\tilde{x} - x$ contains direct information about R_i , and thus about the mechanism shift V .

In contrast, for nonlinear generative models, such information is not linearly decodable, and more sophisticated data augmentations (e.g., learned transformations or contrastive encodings) may be needed to recover V .

Although our synthetic datasets are constructed using linear g , real-world data generally involve nonlinear mechanisms. In these cases, V is not directly observable from $\tilde{x} - x$. Nevertheless, even when unobserved, modelling the latent switch variable V significantly improves performance over baseline implicit models. This underscores the utility of explicitly accounting for soft intervention effects in causal representation learning.

4.3.6 Connections to Prior Work

AICMs are conceptually analogous to *augmented causal networks* introduced in Pearl and Causality (2000), which were originally proposed to model hard interventions. However, as Pearl himself noted:

“One advantage of the augmented network representation is that it is applicable to any change in the functional relationship f_i , and not merely to the replacement of f_i by a constant.” (Pearl & Causality, 2000)

This observation foreshadowed the extension of augmented models to handle *soft interventions*—a direction we pursue with AICMs. Unlike hard interventions, which replace the function entirely, our approach permits nuanced modifications to the functional form, governed by V , enabling causal representation learning in more realistic and weakly controlled settings.

4.4 Identifiability Theorem for Implicit SCMs with Soft Interventions

In this paper, our primary focus is on identifying the causal variables up to reparameterization under soft interventions. To provide a structured presentation, we first provide a concise presentation of the assumptions underlying the identifiability theorem, and next we discuss the role, realism, and robustness of the assumptions highlighting their practical relevance and the extent to which they can be relaxed in real-world scenarios. We then formally define identifiability up to reparameterization (Definition 6) and introduce the **identifiability theorem** (Theorem 1) in the following section. The detailed identifiability assumptions along with the proof of the theorem, is provided in the Appendix 9. and Appendix 10., respectively.

4.4.1 Identifiability Assumptions

The identifiability result relies on the assumptions summarized in Table 3. Formal statements and discussion are provided in Appendix 9..

4.4.2 Assumptions: Role, Realism, and Robustness

The identifiability guarantees rely on a set of structural assumptions that serve two distinct purposes: (i) enabling formal identifiability proofs, and (ii) specifying a well-posed modeling settings. We emphasize that some assumptions-such as Gaussian latent priors, affine structure, and diffeomorphic decoders-are primarily technical conveniences used to make the theoretical analysis tractable, rather than strict requirements for practical implementation.

4.4.2.1 Model-Oriented Assumptions Assumptions such as Gaussianity of the latent variables (Assumption A.8), affinity of the solution functions (Assumption A.9), and diffeomorphic decoders (Assumption A.6) simplify the analytical steps required for identifiability. In particular, Gaussianity allows closed-form arguments in Lemma 1, invertibility, and smoothness prevent pathological information collapse. These conditions ensure uniqueness up to the reparameterization class and avoid degenerate counterexamples. These assumptions may not hold in real-world scenarios and we explore the robustness of our proposed method in these scenarios in real-world experiments conducted in Causal-triplet Liu et al. (2023) datasets presented in Section 6.3.

4.4.2.2 Data-Oriented Assumptions Other assumptions such as atomic interventions (Assumption A.1), known targets (Assumption A.3), sufficient action variability (Assumption A.5), reflect data properties necessary for identifiability under soft interventions. These conditions formalize the intuition that each intervention must induce distinguishable

Table 3 Summary of identifiability assumptions

Assumption	Brief description
A.1: Atomic interventions	Each sample involves an intervention on a single causal variable.
A.2: Complete intervention targets	All causal variables are intervened on in the dataset.
A.3: Known intervention targets	The intervention target is observed for each sample.
A.4: Counterfactual exogenous variables	Only the exogenous variable corresponding to the target changes.
A.5: Sufficient action variability	Interventions induce distinguishable effects in latent space.
A.6: Diffeomorphic decoder	The decoder and solution functions are smooth and invertible.
A.7: Observability of intervention effect	The mechanism shift term can be observed or approximated.
A.8: Gaussian latent variables	Latent causal and exogenous variables follow a multivariate normal distribution.
A.9: Affinity of solution functions	The solution functions satisfy the affine structure described in A.9.

changes in latent space. Similar assumptions appear in prior identifiability results for nonlinear Independent Component Analysis (ICA) and causal representation learning Lachapelle et al. (2022); Ahuja et al. (2023); Brehmer et al. (2022). Furthermore, the results in (Brehmer et al., 2022) suggest that inferring intervention targets is relatively an easy task in implicit causal representation learning, hence, we just assume they are available.

4.4.2.3 Robustness and Sensitivity to Assumption Relaxation Our empirical results demonstrate that the method remains stable when several assumptions are violated in realistic settings particularly in the Causal-triplet experiments presented in Section 6.3. In particular:

- Non-Gaussian latent distributions do not significantly degrade recovery performance in our experiments.
- Neural decoders that are not strictly diffeomorphic still preserve sufficient information for latent alignment.
- Approximate rather than perfectly disjoint supports are sufficient in finite-sample regimes.

This suggests that the assumptions characterize a sufficient theoretical regime for identifiability rather than a necessary one. As commonly observed in nonlinear ICA and causal representation learning, theoretical guarantees often require stronger regularity conditions than those needed in practice.

Furthermore, the proposed method maintains stable performance across multiple encoder architectures (Table 10 and Table 9). We conducted 70 experiments (*five* independent runs per method per dataset) in Table 7 to evaluate the robustness of our method. Our method achieves the lowest standard deviation and the highest mean across metrics over all runs. These observations suggest that moderate violations of the assumptions primarily affect the theoretical guarantees rather than the empirical optimization behaviour.

4.4.2.4 Convergence Considerations Although the identifiability proof relies on the assumptions above, the optimization objective remains well-defined even when these conditions are relaxed. In practice we observe stable training dynamics and consistent convergence across runs, with the loss decreasing smoothly and the recovered representations stabilizing after training, indicating that the learning procedure remains well-behaved in finite-sample regimes.

4.4.3 Identifiability Theorem

We establish identifiability up to reparameterization, allowing for the mapping of causal variables in $\mathcal{Z}$ and $\mathcal{Z}'$ between two Latent Causal Models ($\mathcal{M}$ and $\mathcal{M}'$) through component-wise transformations (Definition 4).

Definition 4 (*Component-wise Transformation*) Let ϕ be a transformation (1-1 onto mapping) between product spaces:

$$\phi : \Pi_{i=1}^n \mathcal{X}_i \rightarrow \Pi_{i=1}^n \mathcal{Y}_i. \quad (15)$$

If there exist local transformations ϕ_i such that:

$$\forall i, j, \forall x, \phi(x_1, x_2, \dots, x_n)_i = \phi_i(x_j), \quad (16)$$

then ϕ is a component-wise transformation.

Definition 5 (*Pushforward measure*) Given a measurable function $f : \mathcal{A} \rightarrow \mathcal{B}$ between two measurable spaces $\mathcal{A}$ and $\mathcal{B}$, and a measure p defined on $\mathcal{A}$, the pushforward measure f_*p on $\mathcal{B}$ is defined for measurable sets E in $\mathcal{B}$ as:

$$(f_*p)(E) = p(f^{-1}(E)), \quad (17)$$

where $*$ denotes the pushforward operation. In other words, the pushforward measure f_*p assigns a measure to a set in $\mathcal{B}$ by measuring the pre-image of that set under f in the space $\mathcal{A}$.

Definition 6 (*Equivalence up to component-wise reparameterization*) Let $\mathcal{M} = (\mathcal{A}, \mathcal{X}, g, \mathcal{I})$ and $\mathcal{M}' = (\mathcal{A}', \mathcal{X}, g', \mathcal{I})$ be two Latent Causal Models (LCM) based on AICMs $\mathcal{A}, \mathcal{A}'$ with shared observation space $\mathcal{X}$, shared intervention targets $\mathcal{I}$, and respective decoders g and g' . We say that $\mathcal{M}$ and $\mathcal{M}'$ are equivalent up to component-wise reparameterization $\mathcal{M} \sim_r \mathcal{M}'$ if there exists a component-wise transformation (Definition 4) $\phi_{\mathcal{Z}}$ from the causal variables in $\mathcal{Z}$ to the causal variables in $\mathcal{Z}'$ and a component-wise transformation $\phi_{\mathcal{E}}$ between $\mathcal{E}$ and $\mathcal{E}'$ such that:

- Indices are preserved (i.e., $\phi_i(z_i) = z'_i$ and $\phi_i(e_i) = e'_i$). Corresponding edges are preserved (i.e., $Z_i \rightarrow Z_j$ holds in G iff $Z'_i \rightarrow Z'_j$ holds in G' . Edges $E_i \rightarrow Z_i$ should be preserved as well.)
- The exogenous transformation preserves the probability measure on exogenous variables $p_{\mathcal{E}'} = (\phi_{\mathcal{E}})_*p_{\mathcal{E}}$ (Definition 5).
- The causal transformation preserves the probability measure on causal variables $p_{\mathcal{Z}'} = (\phi_{\mathcal{Z}})_*p_{\mathcal{Z}}$ (Definition 5).

Theorem 1 (*Identifiability of latent causal models.*) Let $\mathcal{M} = (\mathcal{A}, \mathcal{X}, g, \mathcal{I})$ and $\mathcal{M}' = (\mathcal{A}', \mathcal{X}, g', \mathcal{I})$ be two LCMs with shared observation space $\mathcal{X}$ and shared intervention targets $\mathcal{I}$. Suppose the assumptions in Appendix 9. are satisfied. Then the following statements are equivalent:

1. Two LCMs $\mathcal{M}$ and $\mathcal{M}'$ assign the same likelihood to interventional and observational data i.e., $p_{\mathcal{M}}^{\mathcal{X}}(x, \tilde{x}) = p_{\mathcal{M}'}^{\mathcal{X}}(x, \tilde{x})$.
2. $\mathcal{M}$ and $\mathcal{M}'$ are disentangled, that is $\mathcal{M} \sim_r \mathcal{M}'$ according to Definition 6.

4.5 Training Objective

Consequently, there will be three latent variables in ICRL-SM:

1. A causal mechanism switch variable V .
2. The pre-intervention exogenous variables E .
3. The post-intervention exogenous variables $\tilde{E}$.

As the data log-likelihood $\log p(x, \tilde{x}, \tilde{x} - x) \equiv \log p(x, \tilde{x})$ is intractable, we utilize an ELBO approximation as training objective:

$$\begin{aligned} \log p(x, \tilde{x}) &\geq E_{q(e, \tilde{e}, v|x, \tilde{x})} \left[\log p(x, \tilde{x}|e, \tilde{e}, v) \right] - \text{KLD}(q(e, \tilde{e}, v|x, \tilde{x}) || p(e, \tilde{e}, v)) \\ &= E_{q(v|\tilde{x}-x) \cdot q(e|x) \cdot q(\tilde{e}|\tilde{x})} \left[\log(p(x|e)p(\tilde{x}|\tilde{e})p(\tilde{x}-x|v)) \right] \\ &\quad - \text{KLD}(q(v|\tilde{x}-x) \cdot q(e|x) \cdot q(\tilde{e}|\tilde{x}) || p(\tilde{e}|e, v)p(v)p(e)). \end{aligned} \quad (18)$$

The observations are encoded and decoded independently. The KLD term regularizes the encodings to share the latent *intervention model* $p(\tilde{e}|e, v)p(v)p(e)$ that is shared across all data points. The components of this model can be interpreted as follows:

1. $p(e)$ is the prior distribution over exogenous variables e .
2. $p(v)$ is the prior distribution over switch variables v .
3. $p(\tilde{e}|e, v)$ is a transition model that shows how the exogeneous variables change as a function of the intervention.

We factorize the posterior with a mean-field approximation:

$$q(v, e, \tilde{e}|x, \tilde{x}) = q(v|\tilde{x} - x) \cdot q(e|x) \cdot q(\tilde{e}|\tilde{x}). \quad (19)$$

Following our data generation model (Figure 4), the reconstruction probability is factorized as:

$$p(x, \tilde{x}|e, \tilde{e}, v) = p(x|e)p(\tilde{x}|\tilde{e})p(\tilde{x} - x|v). \quad (20)$$

The prior over latent variables is factorized as (Figure 4):

$$p(\tilde{e}, e, v) = p(\tilde{e}|e, v)p(v)p(e). \quad (21)$$

Pre-intervention exogenous variables are mutually independent, hence:

$$p(e) = \prod_i p(e_i) \quad \text{and} \quad p(v) = \prod_i p(v_i). \quad (22)$$

We assume $p(e_i)$ and $p(v_i)$ to be standard Gaussian. Furthermore, as we assume $e_i = \tilde{e}_i$ for all non-intervened variables, the $p(\tilde{e}|e, v)$ will be as follows:

$$p(\tilde{e}|e, v) = \prod_{i \notin \mathcal{I}} \delta(\tilde{e}_i - e_i) \prod_{i \in \mathcal{I}} p(\tilde{e}_i|e, v) = \prod_{i \notin \mathcal{I}} \delta(\tilde{e}_i - e_i) \prod_{i \in \mathcal{I}} p(\tilde{z}_i|e, v) \left| \frac{\partial \tilde{z}_i}{\partial \tilde{e}_i} \right|. \quad (23)$$

The last equality in Equation 23 is obtained from the Change of Variable Rule in probability theory, applied to the solution function $\tilde{z}_i = s_i(\tilde{e}_i; e_{/i}, v)$. Furthermore, we write $p(\tilde{z}_i|e, v) = p(\tilde{z}_i|e_i)$ since only e_i is a known parent of $\tilde{z}_i$ in implicit modelling. We assume $p(\tilde{z}_i|e_i)$ to be a Gaussian whose mean is determined by e_i .

4.6 Solution Functions

We implement the solution function using a *location-scale noise model (LSNM)* (Immer et al., 2023), following the approach in Brehmer et al. (2022), which ensures that the mapping is a diffeomorphism. Our proposed solution function is defined as:

$$\tilde{z}_i = s_i(\tilde{e}_i; e_{/i}, v) = \frac{\tilde{e}_i - (\text{loc}_i(e_{/i}) + h_i(v))}{\text{scale}_i(e_{/i})}, \quad (24)$$

where:

- $\tilde{z}_i$ denotes the post-intervention latent variable corresponding to the i -th causal factor.
- $\tilde{e}_i$ is the exogenous noise for the i -th latent variable under intervention.
- $e_{/i} = \{e_1, \dots, e_{i-1}, e_{i+1}, \dots, e_n\}$ represents the set of all other exogenous variables in the system except e_i .
- v is the *causal mechanism switch variable*, encoding intervention-induced changes in the structural mechanism.
- $\text{loc}_i : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ is a neural network modeling the location (shift) of $\tilde{z}_i$ as a function of the other exogenous variables $e_{/i}$.
- $\text{scale}_i : \mathbb{R}^{n-1} \rightarrow \mathbb{R}$ is a neural network modeling the scale (variance) of $\tilde{z}_i$ as a function of the other exogenous variables $e_{/i}$.
- $h_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a neural network that captures the influence of the mechanism switch variable v on the i -th latent variable.

The general architecture of the model is illustrated in Figure 3. This formulation explicitly separates the contributions of parental exogenous variables, intervention effects, and mechanism shifts, allowing the model to recover identifiable causal representations under soft interventions.

4.7 Computational Complexity

-
- 1: **Inputs:** Paired pre- and post-intervention samples $(x, \tilde{x}) \sim p(x, \tilde{x})$
 - 2: **Inputs:** Augmented samples $f(\tilde{x}, x)$
 - 3: **Inputs:** Pre-intervention switch variable v_0
 - 4: **Parameters:** Encoders $q(e|x)$, $q(\tilde{e}|\tilde{x})$, and $q(v|f(\tilde{x}, x))$
 - 5: **Parameters:** Decoders $p(x|e)$, $p(\tilde{x}|\tilde{e})$, and $p(f(\tilde{x}, x)|v)$
 - 6: **Parameters:** Structural solution functions $s_i(\tilde{e}_i; e_{/i}, v)$ for $i = 1, \dots, n$
 - 7: **Outputs:** Pre- and post-intervention causal variables $z, \tilde{z}$
 - 8: **Encode observations and mechanism representation:**

$$(\mu_e, \sigma_e) \leftarrow q(e|x)$$

$$(\mu_{\tilde{e}}, \sigma_{\tilde{e}}) \leftarrow q(\tilde{e}|\tilde{x})$$

$$(\mu_v, \sigma_v) \leftarrow q(v|f(\tilde{x}, x))$$
 - 9: **Sample latent variables:**

$$e \sim \mathcal{N}(\mu_e, \sigma_e)$$

$$\tilde{e} \sim \mathcal{N}(\mu_{\tilde{e}}, \sigma_{\tilde{e}})$$

$$v \sim \mathcal{N}(\mu_v, \sigma_v)$$
 - 10: **Compute pre- and post-intervention causal variables:**
 - 11: **for** $i = 1$ to n **do**

$$\tilde{z}_i \leftarrow s_i(\tilde{e}_i; e_{/i}, v)$$

$$z_i \leftarrow s_i(e_i; e_{/i}, v_0)$$
 - 12: **end for**
 - 13: **Reconstruct observations:**

$$\hat{x} \sim p(x|e)$$

$$\hat{\tilde{x}} \sim p(\tilde{x}|\tilde{e})$$

$$\widehat{f(\tilde{x}, x)} \sim p(f(\tilde{x}, x)|v)$$
 - 14: **Maximize the ELBO:**

$$\max \mathcal{L} = \mathbb{E}_{q(v|f(\tilde{x}, x)) q(e|x) q(\tilde{e}|\tilde{x})} \left[\log p(x|e) + \log p(\tilde{x}|\tilde{e}) + \log p(f(\tilde{x}, x)|v) \right]$$

$$- \text{KLD} \left(q(v|f(\tilde{x}, x)) q(e|x) q(\tilde{e}|\tilde{x}) \parallel p(\tilde{e}|e, v) p(v) p(e) \right)$$
 - 15: **Update:** All model parameters are optimized using gradient descent.
-

Algorithm 1 ICRL-SM: Implicit causal representation learning via switchable mechanisms

ICRL-SM is implemented as a standard VAE trained with an ELBO objective. The additional mechanism-switch variable V is inferred from paired pre/post observations (e.g., $\tilde{x} - x$) through a lightweight encoder. Consequently, the computational and memory requirements scale linearly with the latent dimension and batch size and remain comparable to conventional VAE training. The dominant computational cost arises from the forward and backward passes through the encoder and decoder networks. This ensures that the proposed method remains practical and scalable to realistic datasets. The algorithm of our method is also presented at Algorithm 1. In our experiments, we set augmentation function $f(\tilde{x}, x) = \tilde{x} - x$ and pre-intervention switch variable v_0 to vector of zeros. In our real-world experiments for action and object detection, we use pre- and post-intervention causal variables $\tilde{z}$ and z to classify objects and actions in images.

5 Experiments

5.1 Downstream Tasks

5.1.1 Recovering Causal Variables

We validate our theorem using synthetic datasets, assuming Gaussian distribution and linear mixing functions. These serve as a *proof of concept*, where we identify the true causal graph and causal variables from pairs and intervention targets $(x, \tilde{x}, i)$.

5.1.2 Action and Object Inference

We evaluate our model on real-world datasets (Epic-Kitchen, ProcTHOR) from the Causal-triplet Liu et al. (2023) benchmark. These datasets may not meet assumptions like a diffeomorphic decoder, causal variables with Gaussian distribution, and switch variables may not be observable from difference of pre- and post-intervention observations alone. We use a Resnet decoder, and action/object accuracy serves as a proxy for causal variable recovery. Two classifiers predict action and object classes from pre- and post-intervention causal variables. Additionally, we perform an ablation study presented in Section 6.4. Models are trained with known intervention targets, though real-world datasets lack this information. We hypothesize a one-to-one correspondence between causal variables and action types; thus, we assume that the intervention target is given by the action index in the dataset. More details on experiment settings is provided in Appendix 11..

5.2 Datasets

5.2.1 Synthetic Dataset

We generate simple synthetic datasets with $\mathcal{X} = \mathcal{Z} = \mathbb{R}^n$. For each value of n , we generate *ten* random DAGs, a random location-scale SCM, then a random dataset from the parameterized SCM. To generate random DAGs, each edge is sampled in a fixed topological order from a Bernoulli distribution with probability 0.5. The pre-intervention and post-intervention causal variables are obtained as:

$$z_i = \text{scale}(z_{pa_i})e_i + \text{loc}(z_{pa_i}) \xrightarrow{\text{Soft-Intervention}} \tilde{z}_i = \text{scale}(z_{pa_i})\tilde{e}_i + \widetilde{\text{loc}}(z_{pa_i}), \quad (25)$$

where the loc network is changed in post intervention. The pre-intervention loc and post-intervention $\widetilde{\text{loc}}$ network weights are initialized with samples drawn from $\mathcal{N}(0, 1)$ and $\mathcal{N}(3, 1)$, respectively. For ablation studies, we change the mean of these Normal distributions. The scale is constant 1 for both pre-intervention and post-intervention samples. All e_i and $\tilde{e}_i$ are sampled from a standard Gaussian. The causal variables are mapped to the observation space through a randomly sampled $\text{SO}(n)$ rotation. For each dataset, we generate 100 k training samples, 10 k validation samples, and 10 k test samples.

5.2.2 Causal-Triplet

The Causal-triplet datasets (Liu et al., 2023) consist of images containing objects in which an action is manipulating the objects shown in Figure 2. These images include ProcTHOR (Deitke et al., 2022) (100 k image pairs with 7 actions on 24 objects in 10 k environments) and Epic-Kitchens (Damen et al., 2022) (2,632 image pairs of 97 actions). For Epic-Kitchens, we randomly select 20 actions for tractability corresponding to 283 unique objects. Examples of actions and objects in these datasets are given in Table 4.

Based on the actions and objects, we treat causal variables as object attributes that can be modified by actions. Accordingly, actions in these datasets are considered as interventions. For example, let z_1 denote the attributes of an object (e.g., a door); in this case, the target of actions such as opening or closing corresponds to z_1 .

We use action labels in these datasets to identify intervention targets, i.e., to determine which causal variable has been intervened upon. Note that informing the model about the target of intervention is not equivalent to providing the action itself. We use maximum 5 k images from these datasets to train all models, using ResNet encoder–decoder.

5.3 Metrics

For the causal disentanglement task, we are going to use the Disentanglement (D), Completeness (C) Informativeness (I) scores (DCI) (Eastwood & Williams, 2018). According to Eastwood and Williams (2018), Disentanglement (D) and Completeness (C) are sufficient to evaluate the correspondence of learned latents and true causal variables. Therefore, we have utilized the following metrics to evaluate our model.

5.3.1 Causal Disentanglement

Causal disentanglement score quantifies the degree to which the modeled causal variable Z_i factorises or disentangles the ground-truth causal variable Z_i^* . Causal disentanglement D_i for Z_i is calculated as:

$$D_i = (1 - H_K(P_i)) = (1 + \sum_{k=0}^{K-1} P_{ik} \log_K P_{ik}), \quad (26)$$

Table 4 Causal-triplet

(a) ProcTHOR Dataset

Object	Television	Bed	Bed	Television	Laptop	Book	Box
Action	Break	Clean	Dirty	Turn off	Turn on	Open	Close

(b) Epic-Kitchens Dataset

Object	Tofu	Rice	Hob	Bag	Cupboard	Garlic	Tap	Wrap	Rice	Cheese
Action	Insert	Pour	Wash	Fold	Open	Pat	Move	Check	Transition	Stretch
Object	Wrap	Skin	Button	Lid	Plate	Egg	Sponge	Oil	Water	Dough
Action	Flip	Gather	Press	Lock	Wrap	Drop	Water	Carry	Smell	Mark

Examples of actions and objects present in the images of ProcTHOR (a), and Epic-Kitchens (b) datasets

where K is the dimension of Z^* . $P_{ij} = \frac{R_{ij}}{\sum_{k=0}^{K-1} R_{ik}}$ and R_{ij} denotes the probability of Z_i being important for predicting Z_j^* . Total causal disentanglement is the weighted average $\sum_i \rho_i D_i$ where $\rho_i = \frac{\sum_j R_{ij}}{\sum_{ij} R_{ij}}$.

5.3.2 Causal Completeness

Causal Completeness quantifies the degree to which each groundtruth causal variable Z_i^* is captured by a single modeled causal variable Z_i . Causal completeness is calculated as:

$$C_j = (1 - H_D(\tilde{P}_{\cdot j})) = (1 + \sum_{d=0}^{D-1} \tilde{P}_{dj} \log_D \tilde{P}_{dj}), \quad (27)$$

where D denotes the dimension of Z .

5.3.3 Classification Accuracy

DCI (Eastwood & Williams, 2018) scores require ground-truth causal variables, which are not available in the Causal-triplet datasets. Therefore, in our experiments with real-world datasets, we will use classification accuracy as the evaluation metric for the action inference task. As we assume intervention targets are known, we train all models using known intervention targets for a fair comparison.

6 Results and Discussion

6.1 Causal Disentanglement

We generated a dataset for the soft interventions and trained the models of ICRL-SM, ILCM (Brehmer et al., 2022), β -VAE (Higgins et al., 2017), LScale-I (Varici et al., 2025), and dVAE (Locatello et al., 2020) for 10 different seeds, which generated 10 different causal graphs. We selected 4 causal variables to encompass complex causal structures, including forks, chains, and colliders. Table 5 displays the Causal Disentanglement and Causal Completeness scores for all models, computed on the test data.

Note that the proposed model in Locatello et al. (2020) applied an averaging function to the posterior of their causal variables as a substitute to one of their constraints which is not applicable in our setting. Namings used in Locatello et al. (2020) is based on this averaging function. Following Brehmer et al. (2022), we have also used the naming as disentangled VAE (dVAE).

The results in Table 5 show that ICRL-SM identifies the true causal graph in most cases, with the worst performance on graphs $G5$ and $G10$. Causal graphs are usually assumed to be sparse. For $G5$, which is fully connected, the method struggles to identify causal variables. In next experiments, we will investigate factors affecting causal disentanglement, such as the number of edges and soft intervention intensity, which may explain the difficulty in $G10$ despite its sparsity.

Table 5 Synthetic results

Graph	Name	Causal Disentanglement					Causal Completeness				
		β -VAE	dVAE	ILCM	LScale-I	ICRL-SM (ours)	β -VAE	dVAE	ILCM	LScale-I	ICRL-SM (ours)
	G1	0.38	0.54	0.71	0.31	0.82	0.51	0.69	0.78	0.72	0.87
	G2	0.30	0.72	0.75	0.49	0.83	0.49	0.77	0.80	0.63	0.87
	G3	0.28	0.51	0.68	0.63	0.98	0.49	0.56	0.78	0.65	0.98
	G4	0.16	0.50	0.65	0.58	0.68	0.38	0.69	0.77	0.75	0.78
	G5	0.27	0.44	0.53	0.48	0.42	0.45	0.54	0.66	0.48	0.50
	G6	0.52	0.62	0.71	0.40	0.98	0.66	0.69	0.86	0.58	0.98
	G7	0.39	0.49	0.71	0.16	0.75	0.70	0.73	0.89	0.63	0.89
	G8	0.47	0.54	0.50	0.40	0.59	0.6	0.63	0.62	0.60	0.68
	G9	0.30	0.68	0.83	0.32	0.85	0.40	0.76	0.86	0.36	0.87
	G10	0.39	0.39	0.52	0.68	0.32	0.53	0.56	0.82	0.73	0.70

Comparison of identifiability results on synthetic dataset

Table 6 Comparison of ICRL-SM performance on different configurations of $G3$

Edges	Post-intervention causal mechanism	Causal Disentanglement	Causal Completeness
Chain	Default	0.98	0.98
Full	Default	0.89	0.89
Default	Significantly different	0.68	0.73
Default	Almost similar	0.85	0.86

6.2 Factors Affecting Causal Disentanglement

In this experiment, we consider the graph $G3$, which has the best identifiability, and change the intensity of soft intervention and number of edges in its data generation process. To change the intensity, the post-intervention $\widehat{\text{loc}}$ network weights are initialized with samples drawn from $\mathcal{N}(1, 1)$ (almost similar to loc) and $\mathcal{N}(10, 1)$ (significantly different from loc). To change the number of edges, we consider a chain and fully-connected graph. The results in Table 6 further confirms the sparsity of causal graphs as the causal disentanglement is much worse in the fully-connected graph than the default graph of $G3$. The result for significantly different post-intervention causal mechanisms indicate that the switch variable cannot approximate intense effects of soft intervention and more supervision is required to observe V . Similar post-intervention causal mechanisms also do not have sufficient variability to disentangle the causal variables as mentioned in Theorem 1.

6.3 Action and Object Inference

We evaluate ICRL-SM on real-world Causal-triplet datasets. Importantly, several theoretical assumptions used in the identifiability analysis (e.g., Gaussian latent priors and diffeomorphic generative mappings) are not guaranteed to hold in real-world datasets such as Causal-triplet. Nevertheless, the strong empirical performance of ICRL-SM in this setting suggests that the proposed framework remains effective even when these assumptions may not be satisfied in practice. Our results show that explicitly modeling soft intervention effects via V leads to superior performance compared to purely implicit approaches, highlighting the importance of accounting for soft interventions in causal representation learning.

We compare our method against three implicit baselines, β -VAE (Higgins et al., 2017), dVAE (Locatello et al., 2020), and ILCM (Brehmer et al., 2022), as well as explicit causal representation learning methods including ENCO (Lippe et al., 2022) and DDS (Charpentier et al., 2022), together with a fixed-order variant. As shown in Table 7, when predictions are performed using only the intervened causal variable (z_i), ICRL-SM significantly outperforms all baselines, indicating improved disentanglement and more accurate recovery of the underlying causal factors.

For the results shown in Table 7, we perform *five* independent runs with different seeds for each of the *seven* models across *two* datasets, resulting in a total of 70 independent experiments. For each model, we report the mean and standard deviation of the action and object accuracies.

As shown by the results in Table 7, ICRL-SM outperforms all competing methods, achieving higher accuracy and lower standard deviation. The reduced variance indicates the robustness and stability of our proposed model across different runs. In contrast, the higher standard deviations observed for other models suggest limitations in identifiability. Without proper identifiability, these models tend to learn unstable or spurious representations rather than recovering the true underlying causal variables.

Table 7 Action and object accuracy of explicit and implicit models

Model	Method	Epic-Kitchens		ProcTHOR	
		Action Accuracy	Object Accuracy	Action Accuracy	Object Accuracy
Explicit	ENCO (Lippe et al., 2022)	0.62 $\pm$ 0.24	0.06 $\pm$ 0.03	0.67 $\pm$ 0.19	0.51 $\pm$ 0.15
	DDS (Charpentier et al., 2022)	0.57 $\pm$ 0.19	0.05 $\pm$ 0.01	0.51 $\pm$ 0.28	0.52 $\pm$ 0.29
	Fixed-order	0.49 $\pm$ 0.26	0.03 $\pm$ 0.01	0.60 $\pm$ 0.26	0.57 $\pm$ 0.30
Implicit	β -VAE (Higgins et al., 2017)	0.19 $\pm$ 0.01	0.04 $\pm$ 0.01	0.19 $\pm$ 0.02	0.28 $\pm$ 0.14
	dVAE (Locatello et al., 2020)	0.53 $\pm$ 0.06	0.07 $\pm$ 0.02	0.54 $\pm$ 0.21	0.64 $\pm$ 0.32
	ILCM (Brehmer et al., 2022)	0.31 $\pm$ 0.07	0.04 $\pm$ 0.02	0.48 $\pm$ 0.05	0.70 $\pm$ 0.05
	ICRL-SM (ours)	0.69 $\pm$ 0.04	0.07 $\pm$ 0.01	0.86 $\pm$ 0.08	0.78 $\pm$ 0.05

The experiments are conducted using input images with a resolution of 64×64 , which are fed into a ResNet-50 encoder. Predictions are performed using only the intervened causal variable (z_i). Each experiment is repeated over five independent runs with different seeds, and the mean and standard deviation of the accuracies are reported. The best performing method and its results are shown in bold.

6.4 Ablation Studies

6.4.1 Effect of Using Full Causal Representation

To analyze the impact of latent variable usage on prediction performance, we conduct an ablation study in which the full causal representation $\mathbf{z} = (z_1, \dots, z_n)$ is provided to the predictor, i.e., all latent causal variables are used for prediction. In contrast, in our main experiments we use only the intervened causal variable z_i , where the predictor receives solely the latent variable corresponding to the intervened factor, thereby enforcing strict causal isolation.

The rationale behind using only z_i is to ensure that performance gains arise from correctly identifying the true intervened causal factor, rather than from exploiting additional contextual information contained in the remaining latent variables. As shown in Table 8, using the full causal representation generally degrades performance, suggesting that incorporating non-intervened latent factors may introduce spurious correlations that hinders prediction accuracy.

6.4.2 Backbone Model

We trained the models using a simpler backbone model, ResNet-18, to see how it affects performance. The input image resolution is 64×64 and we use the intervened causal variables to predict action and object classes. The results are shown in Table 9. It can be seen from the results that the proposed method outperforms other explicit and implicit models even with a simpler model.

6.4.3 Generalization Across Action Subset

In the main experiments, we evaluate on a randomly selected subset of 20 actions out of the 97 available action categories. To ensure that the observed performance gains are not biased by this particular sampling, we perform an additional ablation study using a different randomly sampled subset of 20 actions corresponding to 277 unique objects. This experiment assesses the sensitivity of the model to the choice of action categories. As shown in Table 10, ICRL-SM outperforms rest of the methods, confirming that the improvements generalize across different action subsets.

Table 8 Comparison of action and object accuracy on the Causal-Triplet datasets under different settings

Method	Epic-Kitchens				ProcTHOR			
	Action Accuracy		Object Accuracy		Action Accuracy		Object Accuracy	
	$\mathbf{z}$	$\mathbf{z_i}$	$\mathbf{z}$	$\mathbf{z_i}$	$\mathbf{z}$	$\mathbf{z_i}$	$\mathbf{z}$	$\mathbf{z_i}$
ICRL-SM	0.16	0.86	0.16	0.18	0.28	0.93	0.40	0.82

Here, $\mathbf{z} = (z_1, \dots, z_n)$ denotes the full set of latent causal variables, while z_i refers to the intervened causal variable only. The notation indicates whether the predictor uses the complete latent representation $\mathbf{z}$ or solely the intervened component z_i for the prediction task. All experiments are conducted using input images with a resolution of 64×64 , and ResNet-50 is adopted as the backbone model. The best performing setting is shown in bold.

Table 9 Action and object accuracy of explicit and implicit models

Model	Method	Epic-Kitchens		ProcTHOR	
		Action Accuracy	Object Accuracy	Action Accuracy	Object Accuracy
Explicit	ENCO (Lippe et al., 2022)	0.14	0.03	0.16	0.28
	DDS (Charpentier et al., 2022)	0.16	0.05	0.34	0.35
	Fixed-order	0.14	0.05	0.34	0.38
Implicit	β -VAE (Higgins et al., 2017)	0.15	0.04	0.20	0.36
	dVAE (Locatello et al., 2020)	0.16	0.02	0.15	0.38
	ILCM (Brehmer et al., 2022)	0.19	0.04	0.15	0.42
	ICRL-SM (ours)	0.35	0.04	0.40	0.69

The experiments are conducted using input images with a resolution of 64×64 , which are fed into a ResNet-18 encoder. Predictions are performed using only the intervened causal variable (z_i). The best performing method and its results are shown in bold.

Table 10 Action and object accuracy of explicit and implicit models

Model	Method	Epic-Kitchens		ProcTHOR	
		Action Accuracy	Object Accuracy	Action Accuracy	Object Accuracy
Explicit	ENCO (Lippe et al., 2022)	0.69	0.13	0.45	0.53
	DDS (Charpentier et al., 2022)	0.44	0.09	0.64	0.67
	Fixed-order	0.79	0.14	0.65	0.54
Implicit	β -VAE (Higgins et al., 2017)	0.18	0.06	0.30	0.37
	dVAE (Locatello et al., 2020)	0.69	0.17	0.81	0.78
	ILCM (Brehmer et al., 2022)	0.59	0.14	0.70	0.76
	ICRL-SM (ours)	0.86	0.18	0.93	0.82

The experiments are conducted using input images with a resolution of 64×64 , which are fed into a ResNet-50 encoder. Predictions are performed using only the intervened causal variable (z_i). In this setting, a different randomly sampled subset of actions is selected from the full set of dataset actions. The best performing method and its results are shown in bold.

6.4.4 Scalability

While our primary research objective centered on addressing identifiability challenges in implicit causal models under soft interventions, we also conducted an investigation into the scalability of our proposed model. To comprehensively assess its performance, we designed experiments covering a range of causal graphs, featuring 5 to 10 variables, with 10 different seeds for each variable, following a similar experimental setup as our 4-variable causal graph experiments. The outcomes of these experiments, comparing ICRL-SM and ILCM, are presented in Figure 5. By increasing the number of variables in the graph, confounding factors and ambiguities of causal relations increase as well. Consequently, more supervision on V is required to better separate the effect of causal variables themselves on the observed variables.

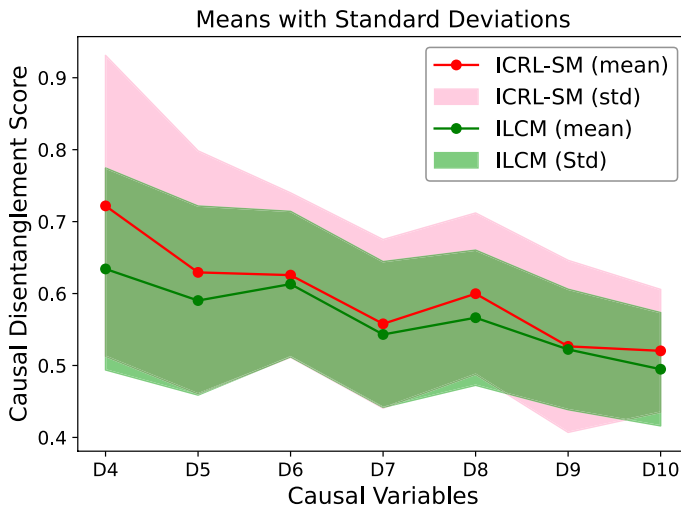


Fig. 5 Causal disentanglement for different number of causal variables

7 Conclusion

In this work, we introduced ICRL-SM, a novel framework for implicit causal representation learning that effectively addresses *soft interventions* by introducing a causal mechanism switch variable. This variable enables the model to capture changes in the underlying causal process induced by such interventions, thereby allowing it to disentangle genuine causal effects from spurious correlations arising from confounding or simultaneous changes in the environment.

Extensive evaluations on both synthetic and real-world image-based datasets demonstrate that ICRL-SM consistently outperforms existing methods. These empirical results strongly support our central hypothesis: explicitly modeling variations in causal mechanisms under soft interventions is critical for learning robust and identifiable causal representations.

Moreover, under a set of well-defined assumptions, we provide theoretical guarantees for the identifiability of the causal model, which are substantiated by our experimental findings.

Overall, ICRL-SM offers a principled approach to causal representation learning in the presence of complex intervention dynamics. Its ability to model soft intervention effects in the data-generating process opens promising directions for future research in causal disentanglement, representation learning, and intervention-aware artificial intelligence. Future extensions may include accommodating more complex generative processes, handling adaptive interventions, or relaxing some of the underlying assumptions in environments.

Appendix

Identifiability Assumptions

Learning identifiable causal representations from soft interventions is inherently ill-posed without suitable assumptions. We introduce a set of assumptions that make identification feasible in our implicit setting and briefly discuss their necessity and potential for relaxation.

Assumption A.1 (*Atomic Interventions*) Let $\mathcal{I} = \{1, 2, \dots, n\}$ denote the intervention target set. Each sample $(x, \tilde{x}, i)$ ($i \in \mathcal{I}$) involves an intervention on only one causal variable Z_i .

- *Rationale*: Ensures attribution of change to a single source.
- *Relaxation*: May be relaxed using multi-target extensions with additional assumptions on sparsity.

Assumption A.2 (*Complete Intervention Target Set*) All causal variables should be intervened on, i.e., $\mathcal{I} = \{1, 2, \dots, n\}$.

- *Rationale*: Guarantees full identifiability coverage.
- *Relaxation*: Identifiability may still hold for partial targets with structural priors or additional data.

Assumption A.3 (*Known Targets*) The target of intervention t is known for each sample.

- *Rationale*: Allows accurate conditioning on the affected variable.
- *Relaxation*: Can be estimated using techniques from Brehmer et al. (2022).

Assumption A.4 (*Counterfactual Exogenous Variables*) Only the exogenous variable corresponding to the target changes:

$$e_j = \tilde{e}_j \quad \forall j \neq i, \quad e_i \neq \tilde{e}_i, \quad i \in \mathcal{I} \quad (28)$$

- *Rationale*: Guarantees only the targeted exogenous variable changes, enabling disjoint support in $(E, \tilde{E})$ and a well-defined intervention map.

Assumption A.5 (*Sufficient Action Variability*) For all distinct intervention targets $i \neq j \in \mathcal{I}$, the distributions of interventional pairs $(z, \tilde{z})$ under i and j are mutually exclusive:

$$\text{supp } p_{\mathcal{M}}^{\mathcal{Z}, \mathcal{I}}(Z, \tilde{Z} \mid I = i) \cap \text{supp } p_{\mathcal{M}}^{\mathcal{Z}, \mathcal{I}}(Z, \tilde{Z} \mid I = j) = \emptyset \quad (29)$$

- *Rationale*: Ensures interventions create distinct, non-overlapping outcomes in latent space for identifiability.

Assumption A.6 (*Diffeomorphic Decoders and Mechanisms*) The decoder g and solution functions s_i are smooth, invertible functions.

- *Rationale*: Prevents information loss and preserves topology.
- *Relaxation*: Empirically, identifiability holds even when this assumption is violated in real datasets.

Assumption A.7 (*Observability of V*) The soft intervention effect term in the Taylor expansion of s_i is observable or can be approximated from $\tilde{x} - x$ when g is linear (More details in Section 4.3).

- *Rationale*: Enables estimation of causal mechanism shifts.
- *Relaxation*: For nonlinear decoders, other augmentations or learned proxies may be used instead. Empirically, identifiability holds even when this assumption is violated in real datasets.

Assumption A.8 (*Gaussianity of Latents*) Causal and exogenous variables follow a multivariate normal distribution.

- *Rationale*: Simplifies identifiability proof in Lemma 1.
- *Relaxation*: Empirically, identifiability holds even when this assumption is violated in real datasets.

Assumption A.9 (*Affinity*) For each i , the solution function $\tilde{s}_i : \mathbb{R} \times \mathbb{R}^{n-1} \times \mathbb{R}^n \rightarrow \mathbb{R}$, defined as $s_i(\tilde{e}_i; e_{/i}, v)$ in Section 4.6, is an affine function of its arguments, which satisfies:

$$s_i(\tilde{e}_i; e_{/i}, v) = h_{1,i}(e_{/i}, v)f_i(\tilde{e}_i) + h_{2,i}(e_{/i}, v), \quad (30)$$

where $f_i : \mathbb{R} \rightarrow \mathbb{R}$ is a diffeomorphism, $h_{1,i} : \mathbb{R}^{n-1} \times \mathbb{R}^n \rightarrow \mathbb{R}$, and $h_{2,i} : \mathbb{R}^{n-1} \times \mathbb{R}^n \rightarrow \mathbb{R}$ are arbitrary functions.

- *Rationale*: Simplifies identifiability proof (Lemma 1).
- *Relaxation*: Empirically, identifiability holds even when this assumption is violated in real datasets.

Identifiability Theorem

Theorem 2 (*Identifiability of latent causal models.*) Let $\mathcal{M} = (\mathcal{A}, \mathcal{X}, g, \mathcal{I})$ and $\mathcal{M}' = (\mathcal{A}', \mathcal{X}, g', \mathcal{I})$ be two LCMs with shared observation space $\mathcal{X}$ and shared intervention targets $\mathcal{I}$. Suppose the assumptions in Appendix 9. are satisfied. Then the following statements are equivalent:

1. Two LCMs $\mathcal{M}$ and $\mathcal{M}'$ assign the same likelihood to interventional and observational data i.e., $p_{\mathcal{M}}^{\mathcal{X}}(x, \tilde{x}) = p_{\mathcal{M}'}^{\mathcal{X}}(x, \tilde{x})$.
2. $\mathcal{M}$ and $\mathcal{M}'$ are disentangled, that is $\mathcal{M} \sim_r \mathcal{M}'$ according to Definition 6.

Proof We will proceed to prove the equivalence between statements 1 and 2 by showing the implication is true in both forward and reverse directions.

Forward Direction: $\mathcal{M} \sim_r \mathcal{M}' \Rightarrow p_{\mathcal{M}}^{\mathcal{X}}(x, \tilde{x}) = p_{\mathcal{M}'}^{\mathcal{X}}(x, \tilde{x})$

This direction is fairly straightforward. According to Definition 6, the fact that $M \sim_r M'$ implies that $\phi_{\mathcal{E}}$ is measure preserving. Thus:

$$p_{\mathcal{M}'}^{\mathcal{E}'}(e', \tilde{e}') = (\phi_{\mathcal{E}})_* p_{\mathcal{M}}^{\mathcal{E}}(e, \tilde{e}) \quad (31)$$

Furthermore, considering that ancestry is preserved, $\phi_{\mathcal{Z}}$ is measure preserving, and that causal variables are obtained from their ancestral exogenous variables in implicit models, we have:

$$p_{\mathcal{M}'}^{\mathcal{Z}'}(z', \tilde{z}') = (\phi_{\mathcal{Z}})_* p_{\mathcal{M}}^{\mathcal{Z}}(z, \tilde{z}) \quad (32)$$

Since models are trained to maximize the log likelihood of $p(x, \tilde{x}, \tilde{x} - x)$ and the latent spaces in M and M' have the same distribution, the decoders should yield the same observational distributions as:

$$p_{\mathcal{M}}^{\mathcal{X}}(x, \tilde{x}) = p_{\mathcal{M}'}^{\mathcal{X}}(x, \tilde{x}) \quad (33)$$

Reverse Direction: $p_{\mathcal{M}}^{\mathcal{X}}(x, \tilde{x}) = p_{\mathcal{M}'}^{\mathcal{X}}(x, \tilde{x}) \Rightarrow \mathcal{M} \sim_r \mathcal{M}'$

Let's define $\phi_{\mathcal{E}} = g'^{-1} \circ g : \mathcal{E} \rightarrow \mathcal{E}'$. Since we can express $e = s^{-1}(z)$, we can now define $\phi_{\mathcal{Z}}$ as:

$$\phi_{\mathcal{Z}} = s' \circ g'^{-1} \circ g \circ s^{-1} : \mathcal{Z} \rightarrow \mathcal{Z}'. \quad (34)$$

Therefore, we have:

$$\phi_{\mathcal{E}} = s'^{-1} \circ \phi_{\mathcal{Z}} \circ s \quad (35)$$

Because g and g' are diffeomorphisms (Assumption A.6), $\phi_{\mathcal{E}}$ is a diffeomorphism as well. Furthermore, since $p_{\mathcal{M}}^{\mathcal{X}} = p_{\mathcal{M}'}^{\mathcal{X}}$ and $\phi_{\mathcal{E}}$ is a diffeomorphism, then:

$$p_{\mathcal{M}'}^{\mathcal{E}'}(e', \tilde{e}') = (\phi_{\mathcal{E}})_* p_{\mathcal{M}}^{\mathcal{E}}(e, \tilde{e}) \quad (36)$$

Consequently, $\phi_{\mathcal{E}}$ is measure-preserving. Similarly, $\phi_{\mathcal{Z}}$ is measure-preserving as well since causal mechanisms are diffeomorphic (Assumption A.6).

Step 1: Identical Correspondence of Edges and Nodes

We assume interventions are atomic in Assumption A.1. Let's define the set U for atomic interventions $i \neq j \in \mathcal{I}$ as:

$$U = \{(e, \tilde{e}) | \text{supp } p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(E, \tilde{E} | I = i) \cap \text{supp } p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(E, \tilde{E} | I = j)\} \subset \mathcal{E} \times \mathcal{E} \quad (37)$$

By Assumption A.4, for any pair $(e, \tilde{e}) \sim p(E, \tilde{E} \mid I = i)$, only the component e_i changes between e and $\tilde{e}$ while all others remain fixed: $e_k = \tilde{e}_k \quad \forall k \neq i$ and $e_i \neq \tilde{e}_i$.

Similarly, for $j \neq i$, any sample from $p(E, \tilde{E} \mid I = j)$ satisfies $e_k = \tilde{e}_k \quad \forall k \neq j$, and $e_j \neq \tilde{e}_j$.

Thus, any pair $(e, \tilde{e})$ from $\text{supp } p(E, \tilde{E} \mid I = i)$ must satisfy $e_i \neq \tilde{e}_i$, but $e_j = \tilde{e}_j$, while pairs from $\text{supp } p(E, \tilde{E} \mid I = j)$ satisfy the reverse. Therefore, no pair can simultaneously satisfy both constraints, and the supports are disjoint:

$$\begin{aligned} \text{supp } p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(E, \tilde{E} \mid I = i) \cap \text{supp } p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(E, \tilde{E} \mid I = j) &= \emptyset \\ \Rightarrow U &= \emptyset \\ \Rightarrow p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(U \mid I = i) &= p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(U \mid I = j) = 0 \end{aligned} \quad (38)$$

Under Assumption A.2, we can write the marginal distribution $p_{\mathcal{M}}^{\mathcal{E}}(E, \tilde{E})$ as:

$$p_{\mathcal{M}}^{\mathcal{E}}(E, \tilde{E}) = \sum_{i \in I} p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(E, \tilde{E} \mid I = i) p_{\mathcal{M}}^{\mathcal{I}}(I = i), \quad (39)$$

According to Equation 38, we can say that $p_{\mathcal{M}}^{\mathcal{E}}(E, \tilde{E})$ is a discrete mixture of non-overlapping distributions $p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(E, \tilde{E} \mid I = i)$. Similarly, we can say that $p_{\mathcal{M}'}^{\mathcal{E}'}(E', \tilde{E}')$ is a discrete mixture of non-overlapping distributions. Since $\phi_{\mathcal{E}}$ is measure preserving, $p_{\mathcal{M}}^{\mathcal{E}}(E, \tilde{E}) = p_{\mathcal{M}'}^{\mathcal{E}'}(E', \tilde{E}')$. Therefore,

$$\sum_{i \in I} p_{\mathcal{M}}^{\mathcal{E}, \mathcal{I}}(E, \tilde{E} \mid I = i) p_{\mathcal{M}}^{\mathcal{I}}(I = i) = \sum_{i' \in I'} p_{\mathcal{M}'}^{\mathcal{E}', \mathcal{I}'}(E', \tilde{E}' \mid I' = i') p_{\mathcal{M}'}^{\mathcal{I}'}(I' = i'), \quad (40)$$

It can be concluded that as $\phi_{\mathcal{E}}$ must map between the conditional distributions, there exists a bijection that also induces a permutation $\psi : [n] \rightarrow [n]$.

Under Assumption A.3, we have a shared domain $\mathcal{I}$ in $\mathcal{M}$ and $\mathcal{M}'$. Because the mixture over $(E, \tilde{E})$ is equal across models (Equation 40), the conditional components must match under the same index. Therefore, the permutation ψ is the identity transformation.

In the latent space $\mathcal{Z}$, it is crucial that interventions induce sufficient variability so that the conditional interventional distributions $p_{\mathcal{M}}^{\mathcal{Z}, \mathcal{I}}(Z, \tilde{Z} \mid I = i)$ are disjoint for different intervention targets i .

Because soft interventions modify the underlying causal mechanisms rather than directly fixing variables, the value of causal variable $\tilde{z}$ depends on all ancestral exogenous variables affecting the intervened variable. Since these exogenous variables may influence multiple causal variables, insufficiently distinct mechanism changes can cause overlaps in the distributions associated with different interventions. This overlap arises because similar shifts in ancestral variables produce similar latent outcomes, blurring the distinctions between interventions.

By assuming sufficient variability (Assumption A.5), we ensure that each intervention $i \in \mathcal{I}$ produces a unique, non-overlapping distribution in $\mathcal{Z} \times \mathcal{Z}$. This property mirrors the non-overlapping supports we have assumed in the exogenous space $\mathcal{E}$.

Consequently, there exists a permutation mapping between the interventional distributions $p_{\mathcal{M}}^{\mathcal{Z}, \mathcal{I}}(\mathcal{Z}, \tilde{\mathcal{Z}} \mid I = i)$ and $p_{\mathcal{M}'}^{\mathcal{Z}', \mathcal{I}}(\mathcal{Z}', \tilde{\mathcal{Z}}' \mid I = i)$. Since the intervention targets are known and shared, this permutation reduces to the identity mapping, preserving the correspondence between interventions across models.

The effect of soft intervention with known targets on these conditional distributions is shown in Figure 6.

Step 2: Component-Wise $\phi_{\mathcal{Z}}$

We are going to propose a Lemma and prove it before proceeding to the next step in a proof.

Lemma 1 *The transformation $\phi_{\mathcal{Z}} : \mathcal{Z} \rightarrow \mathcal{Z}'$ between the causal variable of two LCMs $\mathcal{M}$ and $\mathcal{M}'$ defined in Definition 6 is a component-wise transformation, if $\forall i \neq j \quad \tilde{E}_i' \perp\!\!\!\perp \tilde{E}_j' \mid E', V'$.*

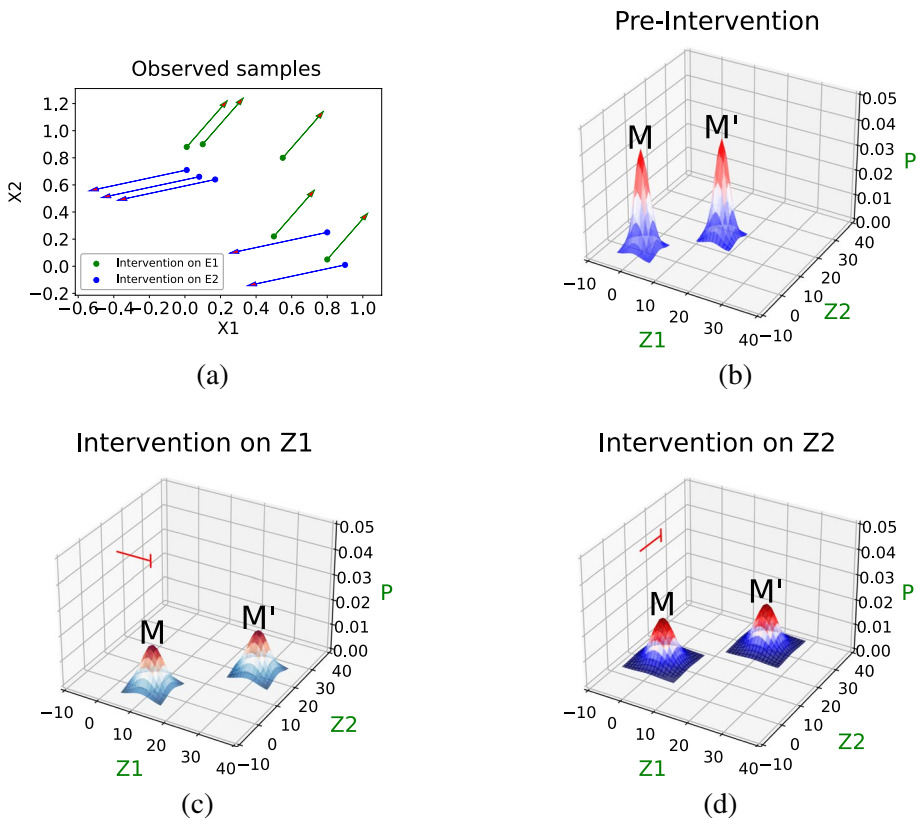


Fig. 6 The distribution of observed and causal variables in two causal models $\mathcal{M}$ and $\mathcal{M}'$, which belong to the equivalence class up to reparameterization. (a) There are 10 observed samples in which Z_1 or Z_2 has been intervened on. (b) The distribution of causal variables when $I = 0$ (no intervention) is identical to each other but the range of value of causal variables are different and can be mapped to each other using $\phi_{\mathcal{Z}}$. (c) The intervention on Z_1 ($I = 1$). (d) The intervention on Z_2 ($I = 2$). For $I = 1$ and $I = 2$ the distributions are again identical to each other but are different for different targets of intervention as soft interventions change the conditional distribution (condition on parents) of causal variables. Also, for each value of I , the distributions of $\mathcal{M}$ and $\mathcal{M}'$ should move in one direction as targets are known

proof:

Under Assumption A.9 and Assumption A.7 (V is observed and shared between $\mathcal{M}$ and $\mathcal{M}'$) we can write:

$$\tilde{z}'_i = s'_i(\tilde{e}'_i; e'_{/i}, v) = h'_{1,i}(e'_{/i}, v) \cdot f'_i(\tilde{e}'_i) + h'_{2,i}(e'_{/i}, v)$$

for simplicity let's just say $\tilde{Z}'_i = h'_{1,i} \cdot f'_i(\tilde{e}'_i) + h'_{2,i}$.

Given $\tilde{E}'_i \perp\!\!\!\perp \tilde{E}'_j \mid E', V$, we compute:

$$\begin{aligned} \mathbb{E}[\tilde{Z}'_i \tilde{Z}'_j \mid E' = e', V = v] &= \mathbb{E}[(h'_{1,i} f'_i(\tilde{E}'_i) + h'_{2,i})(h'_{1,j} f'_j(\tilde{E}'_j) + h'_{2,j})] \\ &= h'_{1,i} h'_{1,j} \mathbb{E}[f'_i(\tilde{E}'_i)] \mathbb{E}[f'_j(\tilde{E}'_j)] + h'_{1,i} h'_{2,j} \mathbb{E}[f'_i(\tilde{E}'_i)] \\ &\quad + h'_{1,j} h'_{2,i} \mathbb{E}[f'_j(\tilde{E}'_j)] + h'_{2,i} h'_{2,j}, \end{aligned}$$

Similarly:

$$\begin{aligned} \mathbb{E}[\tilde{Z}'_i \mid E' = e, V = v] &= h'_{1,i} \mathbb{E}[f'_i(\tilde{E}'_i)] + h'_{2,i}, \\ \mathbb{E}[\tilde{Z}'_j \mid E' = e, V = v] &= h'_{1,j} \mathbb{E}[f'_j(\tilde{E}'_j)] + h'_{2,j}, \end{aligned}$$

and thus,

$$\mathbb{E}[\tilde{Z}'_i \mid E', V] \mathbb{E}[\tilde{Z}'_j \mid E', V] = \mathbb{E}[\tilde{Z}'_i \tilde{Z}'_j \mid E', V].$$

Hence:

$$\text{Cov}(\tilde{Z}'_i, \tilde{Z}'_j \mid E', V) = 0$$

Now, under Assumption A.8 we can write:

$$\text{Cov}(\tilde{Z}'_i, \tilde{Z}'_j \mid E', V) = 0 \Rightarrow \tilde{Z}'_i \perp\!\!\!\perp \tilde{Z}'_j \mid E', V$$

Let $\phi_{\mathcal{E}} = g'^{-1} \circ g : \mathcal{E} \rightarrow \mathcal{E}'$, where g and g' are the decoders in $\mathcal{M}$ and $\mathcal{M}'$, respectively.

As per Assumption A.6, both decoders are diffeomorphic, so $\phi_{\mathcal{E}}$ is also a diffeomorphism.

Let $s(\tilde{e}; e, v) = [s_1(\tilde{e}_1; e_{/1}, v), s_2(\tilde{e}_2; e_{/2}, v), \dots, s_n(\tilde{e}_n; e_{/n}, v)]$ denote the set of all solution functions in the post-intervention system, which are diffeomorphic by Assumption A.6. Furthermore, let's define $\tilde{Z} = (\tilde{Z}_1, \dots, \tilde{Z}_n) \in \mathbb{R}^n$. Then:

$$\begin{aligned}
(\phi_{\mathcal{E}}^{-1} \text{ is diffeomorphic}) &\Rightarrow \tilde{Z}'_i \perp\!\!\!\perp \tilde{Z}'_j \mid \phi_{\mathcal{E}}^{-1}(E'), V \\
&\Rightarrow \tilde{Z}'_i \perp\!\!\!\perp \tilde{Z}'_j \mid E, V \\
(\text{definition of conditional independence}) &\Rightarrow p(\tilde{Z}'_i \mid E = e, V = v), p(\tilde{Z}'_j \mid E = e, V = v) \\
&= p(\tilde{Z}'_i, \tilde{Z}'_j \mid E = e, V = v) \\
(\tilde{Z} \text{ are deterministic maps of } \tilde{E}, E, \text{ and } V) &\Rightarrow p(\tilde{Z}'_i \mid \tilde{E}, E = e, V = v), p(\tilde{Z}'_j \mid \tilde{E}, E = e, V = v) \\
&= p(\tilde{Z}'_i, \tilde{Z}'_j \mid \tilde{E}, E = e, V = v) \\
(s_i \text{ are diffeomorphic}) &\Rightarrow p(\tilde{Z}'_i \mid s(\tilde{e}; e, v), E = e, V = v), p(\tilde{Z}'_j \mid s(\tilde{e}; e, v), E = e, V = v) \\
&= p(\tilde{Z}'_i, \tilde{Z}'_j \mid s(\tilde{e}; e, v), E = e, V = v) \\
&\Rightarrow p(\tilde{Z}'_i \mid \tilde{Z} = \tilde{z}, E = e, V = v), p(\tilde{Z}'_j \mid \tilde{Z} = \tilde{z}, E = e, V = v) \\
&= p(\tilde{Z}'_i, \tilde{Z}'_j \mid \tilde{Z} = \tilde{z}, E = e, V = v) \\
&\Rightarrow \tilde{Z}'_i \perp\!\!\!\perp \tilde{Z}'_j \mid \tilde{Z}, E, V
\end{aligned}$$

We use the Pigeonhole principle to arrive at the conclusion. Let's define holes as $\tilde{Z} = (\tilde{Z}_1, \dots, \tilde{Z}_n) \in \mathbb{R}^n$ and pigeons as the dependencies of $\tilde{Z}' = (\tilde{Z}'_1, \dots, \tilde{Z}'_n) \in \mathbb{R}^n$ - for each i , let the $\tilde{Z}'_i$ depend on some subset of $\tilde{Z}$. So if $\tilde{Z}'_i$ depends on $k_i \geq 1$ number of variables in $\tilde{Z}$, we say that it occupies k_i holes. The reason for $k_i \geq 1$ is that $\mathcal{M}$ and $\mathcal{M}'$ have shared observation space.

Suppose for contradiction that at least one $\tilde{Z}'_i$ depends on two or more variables $\tilde{Z}$; say $\tilde{Z}_j$ and $\tilde{Z}_k$, with $j \neq k$. So $k_i \geq 2$.

To satisfy the conditional independence $\tilde{Z}'_i \perp\!\!\!\perp \tilde{Z}'_r \mid \tilde{Z}, E, V$, no two $\tilde{Z}'_i, \tilde{Z}'_r$ can share any common dependency on the same variable from $\tilde{Z}$; because shared dependency creates conditional dependence given $\tilde{Z}$, violating the assumption.

Now we apply the pigeonhole principle:

There are n holes and there are n pigeons, each requiring at least one hole. At least one $\tilde{Z}'_i$ requires at least two holes. In order to not violate the assumption, we would need more than n holes in total which is a contradiction. Hence, $\phi_{\mathcal{Z}}$ must be component-wise.

According to Lemma 1, in order to prove that $\phi_{\mathcal{Z}}$ is a component-wise transformation, we need to prove that $\forall i \neq j \quad \tilde{E}'_i \perp\!\!\!\perp \tilde{E}'_j \mid E', V'$. In implicit modeling we do not know the parents of each causal variable. Since E'_i is a known parent of $\tilde{Z}'_i$. The mean of the conditional distribution of $p_{\mathcal{M}'}^{\tilde{Z}}(\tilde{Z}'_i \mid E' = e', V' = v')$ can be calculated as:

$$\mathbb{E}[\tilde{Z}'_i \mid E' = e', V' = v'] = \mu_{\tilde{Z}'_i} + \rho \frac{\sigma_{\tilde{Z}'_i}}{\sigma_{E'_i}} (e'_i - \mu_{E'_i}) \quad (41)$$

where ρ and σ are the correlation coefficient and variance of the random variables, respectively. On the other hand, we model $\tilde{Z}'_i$ using switch mechanisms as:

$$\tilde{z}'_i = s_i(\tilde{e}'_i; e'_{/i}, v') \quad (42)$$

By using Taylor's expansion we can write Equation 42 as:

$$\begin{aligned}
s_i(\tilde{e}'_i; e'_{/i}, v') &= s_i(\tilde{e}'_i; e'_{/i}, v'_0) + \sum_{n=1}^{\infty} \frac{1}{n!} \left(\frac{\partial^n s_i}{\partial V^n} \bigg|_{V'=v'_0} (v' - v'_0)^n \right) \\
&= s_i(\tilde{e}'_i; e'_{/i}, v'_0) + R_i
\end{aligned} \quad (43)$$

Consequently, we can write the following equality from Equation 41:

$$\mathbb{E}[\tilde{Z}'_i | E' = e', V' = v'] = \mathbb{E}[s'_i(\tilde{E}'_i; e'_{/i}, v'_0) + R_i] = \mu_{\tilde{Z}'_i} + \rho \frac{\sigma_{\tilde{Z}'_i}}{\sigma_{E'_i}} (e'_i - \mu_{E'_i}) \quad (44)$$

Under hard intervention, structural equations are replaced by constants; thus, the right-hand side contains no dependence on ancestor variables. This implies:

$$\forall i \neq j \quad \tilde{E}'_i \perp\!\!\!\perp \tilde{E}'_j$$

In contrast, under soft intervention, parent variables can still influence $\tilde{Z}'_i$ through s'_i , which introduces dependencies if we did not have the R_i term:

$$\exists i \neq j \quad \tilde{E}'_i \not\perp\!\!\!\perp \tilde{E}'_j$$

By introducing the causal mechanism switch variable V' , and assuming it is observed, we isolate the soft intervention effects into an additive term R_i . This enables the model to learn exogenous variables $\tilde{E}'_i$ that satisfy:

$$\forall i \neq j \quad \tilde{E}'_i \perp\!\!\!\perp \tilde{E}'_j \mid E', V',$$

which is consistent with the standard assumption that exogenous variables are mutually independent. Consequently:

$$\forall i \neq j \quad \tilde{E}'_i \perp\!\!\!\perp \tilde{E}'_j \mid E', V' \rightarrow \phi_{\mathcal{Z}} \text{ is a component-wise transformation.}$$

Step 3: Component-Wise $\phi_{\mathcal{E}}$

Using the result from previous step that $\phi_{\mathcal{Z}}$ is a component-wise transformation, the string diagrams for connections between E and E' will be as shown in Figure 7. $\phi_{\mathcal{E}}(e)_{i'} = (s'^{-1} \circ \phi_{\mathcal{Z}} \circ s)(e)_{i'}$ will only depend on E_{anc_i} , where anc_i is the indices of ancestors of variable i .

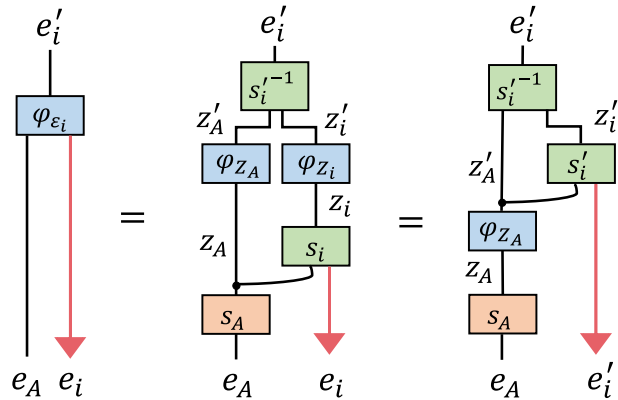
Because:

$$\begin{aligned} & (\text{permutations of indices is identity}) \rightarrow \phi_{\mathcal{E}}(e)_{i'} = \phi_{\mathcal{E}}(e)_i \\ & (s'^{-1} \text{ only depends on } z'_i \text{ and } z'_{anc_i}) \rightarrow \phi_{\mathcal{E}}(e)_i = s'^{-1}_i(z'_i, z'_{anc_i}) \\ & (\phi_{\mathcal{Z}} \text{ is a component-wise transformation}) \rightarrow s'^{-1}_i(z'_i, z'_{anc_i}) = s'^{-1}_i(\phi_{\mathcal{Z}}(z_i)_i, \phi_{\mathcal{Z}}(z_{anc_i})_{anc_i}) \\ & (s_i \text{ and } s_{anc_i} \text{ only depend on } e_i \text{ and } e_{anc_i}) \rightarrow s'^{-1}_i(\phi_{\mathcal{Z}}(z_i)_i, \phi_{\mathcal{Z}}(z_{anc_i})_{anc_i}) = \\ & s'^{-1}_i(\phi_{\mathcal{Z}}(s_i(e_i, e_{anc_i}))_i, \phi_{\mathcal{Z}}(s_{anc_i}(e_{anc_i}))_{anc_i}) \end{aligned}$$

Please note that the arguments of the solution functions presented above reflect their dependencies, rather than the precise inputs used in the implementation.

The first equality in Figure 7 follows from the definition of $\phi_{\mathcal{E}_i}$. The second equality holds when we first apply $\phi_{\mathcal{Z}_A}$ and then apply the causal mechanisms. It can be concluded from

Fig. 7 String diagrams for connections between E and E' . The triangle indicates sampling variables from their corresponding distributions



the most right-hand side diagram in Figure 7 that the transformation from $\mathcal{E}_i \times \mathcal{E}_A \rightarrow \mathcal{E}'_i$ is constant in $\mathcal{E}_A$. Therefore, $\phi_{\mathcal{E}_i}$ is a component-wise transformation.

Experiments

This section contains additional details about ICRL-SM design architectures and experiments settings.

Architecture Design

Based on the ICRL-SM architecture (Figure 3), we design a location-scale solution function (Equation 24), where the loc_i , scale_i , and h_i networks are all fully connected. Each network consists of two layers with 64 hidden units per layer and ReLU activations. The encoder and decoder for latents E and $\bar{E}$ are shared, while a separate encoder-decoder pair with the same architecture is used for latent V .

For synthetic dataset experiments, encoders and decoders are fully connected networks with two hidden layers of 64 units each. For the Causal-triplet datasets, we employ ResNet backbones. All baseline models use the same encoder-decoder architectures to ensure fair comparison. Specifically:

ResNet-50 experiments (Table 7, Table 10): ResNet-50 encoder and decoder, with classifiers having one hidden layer of 64 units, predict actions and objects.

ResNet-18 experiments (Table 9): ResNet-18 encoder and decoder, with classifiers of two hidden layers and 2 units each, predict actions and objects.

Training

To enforce the condition described in Equation 23 for $i \notin \mathcal{I}$, we assign the post-intervention exogenous variables the same value as the pre-intervention exogenous variables. In mathematical terms, this translates to $\forall i \notin \mathcal{I}$, we set $\bar{e}_i = e_i$.

In our experiments, we do not pretrain the networks, however, for the baseline models we follow the training procedure in Brehmer et al. (2022). In addition, we impose a con-

sistency constraint to encourage the encoder and decoder to act as approximate inverses of each other. The consistency regularizer is defined as:

$$\sum_i \mathbb{E}_{x \sim p(x)} \mathbb{E}_{\hat{x} \sim p(\hat{x}|e)} [(x_i - \hat{x}_i)^2], \quad (45)$$

where $\hat{x}$ denotes the reconstructed samples produced by the decoder.

Batch Size. In all experiments, the batch size is 64.

Learning Rate. For optimization, we use the Adam optimizer with default hyperparameters. In synthetic experiments (Table 5), the learning rate decays from 3×10^{-4} to 1×10^{-8} using a cosine scheduler. For the Causal-triplet experiments (Table 7), the learning rate decays from 2×10^{-3} to 1×10^{-8} , while for the ResNet-18 experiments (Table 9) it decays from 1×10^{-4} to 1×10^{-8} .

Epoch. Training schedules are 400 epochs for Table 7, 2000 epochs for Table 9, and 100 epochs for the synthetic experiments (Table 5). For Table 9, graph parameters of the explicit models are frozen after 1000 epochs.

Compute Infrastructure. All models are trained on NVIDIA GeForce RTX 4090 GPUs. Each Causal-Triplet experiment requires 3–8 hours, while each synthetic experiment requires 2–3 hours. We save the model weights with the best validation loss and use them for evaluation on the test data.

Author Contributions SSGB developed and proved the theoretical framework. With guidance from ZG, SSGB designed, implemented the experiments, and analyzed the results. ZG generated all figures and tables. SSGB and ZG jointly wrote the manuscript. OS contributed to the theoretical proofs. OS and MC reviewed and revised the manuscript. SSGB and ZG jointly revised the manuscript and responded to the reviewers comments.

Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

References

- Ahuja, K., Mahajan, D., Wang, Y., & Bengio, Y. (2023). Interventional causal representation learning. Proceedings of Machine Learning Research. *International Conference on Machine Learning, ICML* (Vol. 202, pp. 372–407). PMLR.
- Bagi, S. S. G., Gharace, Z., Schulte, O., & Crowley, M. (2023). Generative causal representation learning for out-of-distribution motion forecasting. Proceedings of Machine Learning Research. *International Conference on Machine Learning, ICML* (Vol. 202, pp. 31596–31612). PMLR.
- Brehmer, J., De Haan, P., Lippe, P., & Cohen, T. S. (2022). Weakly supervised causal representation learning. *Advances in Neural Information Processing Systems*, 35, 38319–38331.
- Buchholz, S., Rajendran, G., Rosenfeld, E., Aragam, B., Schölkopf, B., & Ravikumar, P. (2023). Learning linear causal representations from interventions under general nonlinear mixing. *Advances in Neural Information Processing Systems*, 36, 45419–62.
- Charpentier, B., Kibler, S., & Günnemann, S. (2022). Differentiable DAG sampling. In: *The Tenth International Conference on Learning Representations, ICLR* April 25–29. OpenReview.net, Virtual Event.
- Cooper, G. F., & Yoo, C. (2013). Causal Discovery from a Mixture of Experimental and Observational Data

- Correa, J. D., Bareinboim, E., Larochelle, H., Ranzato, M., Hadsell, R., & Balcan, M. (2020). General transportability of soft interventions: Completeness results. In H. Lin (Ed.), *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems*. NeurIPS
- Damen, D., Doughty, H., Farinella, G. M., Furnari, A., Kazakos, E., Ma, J., Moltisanti, D., Munro, J., Perrett, T., Price, W., & Wray, M. (2022). Rescaling egocentric vision: Collection, pipeline and challenges for EPIC-KITCHENS-100. *International Journal of Computer Vision*, 130(1), 33–55. <https://doi.org/10.1007/s11263-021-01531-2>
- Deitke, M., VanderBilt, E., Herrasti, A., Weihs, L., Ehsani, K., Salvador, J., Han, W., Kolve, E., Kembhavi, A., & Mottaghi, R. (2022). Procthor: Large-scale embodied ai using procedural generation. *Advances in Neural Information Processing Systems*, 35, 5982–5994.
- Eastwood, C., & Williams, C. K. I. (2018). A framework for the quantitative evaluation of disentangled representations. 6th International Conference on Learning Representations, ICLR
- Higgins, I., Matthey, L., Pal, A., Burgess, C. P., Glorot, X., Botvinick, M. M., Mohamed, S., & Lerchner, A. (2017). beta-vae: Learning basic visual concepts with a constrained variational framework. *5th International Conference on Learning Representations*, ICLR
- Immer, A., Schulteis, C., Vogt, J. E., Schölkopf, B., Bühlmann, P., & Marx, A. (2023). On the identifiability and estimation of causal location-scale noise models. *Proceedings of Machine Learning Research. International Conference on Machine Learning, ICML 2023, 23–29 July* (Vol. 202, pp. 14316–14332). PMLR.
- Jaber, A., Kocaoglu, M., Shanmugam, K., & Bareinboim, E. (2020). Causal discovery from soft interventions with unknown targets: Characterization and learning. *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems*. NeurIPS
- Kaddour, J., Lynch, A., Liu, Q., Kusner, M. J., & Silva, R. (2022). Causal machine learning: A survey and open problems. *arXiv preprint arXiv:2206.15475*.
- Kivva, B., Rajendran, G., Ravikumar, P. K., & Aragam, B. (2022). Identifiability of deep generative models under mixture priors without auxiliary information. *UAI 2022 Workshop on Causal Representation Learning*
- Kügelgen, J., Besserve, M., Wendong, L., Gresele, L., Kekić, A., Bareinboim, E., Blei, D. M., & Schölkopf, B. (2023). Nonparametric identifiability of causal representations from unknown interventions. *Advances in Neural Information Processing Systems*, 36, 48603–38.
- Kügelgen, J., Sharma, Y., Gresele, L., Brendel, W., Schölkopf, B., Besserve, M., & Locatello, F. (2021). Self-supervised learning with data augmentations provably isolates content from style. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P. S. Liang, & J. W. Vaughan (Eds.), *Advances in Neural Information Processing Systems* (Vol. 34, pp. 16451–16467). Curran Associates, Inc.
- Lachapelle, S., Rodriguez, P., Sharma, Y., Everett, K., Priol, R.L., & Lacoste, A., Lacoste-Julien, S. (2022). Disentanglement via mechanism sparsity regularization: A new principle for nonlinear ICA. In: 1st Conference on Causal Learning and Reasoning, CLear. *Proceedings of Machine Learning Research*, vol. 177, pp. 428–484. PMLR, Sequoia Conference Center.
- Li, R., Pu, Y., Li, Z., Wu, C., Xie, H., & Lian, D. (2025). Invariant representation learning via decoupling style and spurious features. *Machine Learning*, 114(2), 37.
- Lippe, P., Cohen, T., & Gavves, E. (2022). Efficient neural causal discovery without acyclicity constraints. In: *The Tenth International Conference on Learning Representations*, ICLR April 25–29. OpenReview.net, Virtual Event.
- Lippe, P., Magliacane, S., Löwe, S., Asano, Y. M., Cohen, T., & Gavves, S. (2022). CITRIS: causal identifiability from temporal intervened sequences. *Proceedings of Machine Learning Research. International Conference on Machine Learning, ICML* (Vol. 162, pp. 13557–13603). PMLR.
- Liu, C., Sun, X., Wang, J., Tang, H., Li, T., Qin, T., Chen, W., & Liu, T.-Y. (2021). Learning causal semantic representation for out-of-distribution prediction. *NeurIPS 2021, December 6–14. Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021* (Vol. 34, pp. 6155–6170). Virtual Event: Curran Associates Inc.
- Liu, Y., Alahi, A., Russell, C., Horn, M., Zietlow, D., Schölkopf, B., & Locatello, F. (2023). Causal triplet: An open challenge for intervention-centric causal representation learning. *Proceedings of Machine Learning Research. Conference on Causal Learning and Reasoning, CLear 2023, 11–14 April* (Vol. 213, pp. 553–573). Amazon Development Center. PMLR.
- Liu, Y., Cadei, R., Schweizer, J., Bahmani, S., & Alahi, A. (2022). Towards robust and adaptive motion forecasting: A causal representation perspective. In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, June 18–24, 2022*, pp. 17060–17071. IEEE. <https://doi.org/10.1109/CVPR52688.2022.01657>

- Locatello, F., Poole, B., Rätsch, G., Schölkopf, B., Bachem, O., & Tschannen, M. (2020). Weakly-supervised disentanglement without compromises. In: *Proceedings of the 37th International Conference on Machine Learning*, ICML 2020, 13–18 July 2020. Proceedings of Machine Learning Research, vol. 119, pp. 6348–6359. PMLR, Virtual Event.
- Lu, C., Wu, Y., Hernández-Lobato, J. M., & Schölkopf, B. (2022). Invariant causal representation learning for out-of-distribution generalization. The Tenth International Conference on Learning Representations, ICLR
- Ni, H., Wang, T.-Z., Tao, H., Huang, X., & Hou, C. (2025). Gradient-based causal discovery with latent variables. *Machine Learning*, 114(2), 46.
- Pearl, J. (2005). Causality, cambridge university press (2000). Artificial Intelligence. 169(2), 174–179.
- Pearl, J., Glymour, M., & Jewell, N. P. (2016). *Causal Inference in Statistics: A Primer* (2nd ed.). John Wiley & Sons.
- Perry, R., Kügelgen, J., & Schölkopf, B. (2022). Causal discovery in heterogeneous environments under the sparse mechanism shift hypothesis. In: Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022.
- Schölkopf, B. (2022). *Probabilistic and Causal Inference: The Works of Judea Pearl* (pp. 765–804) Causality for machine learning.
- Schölkopf, B., Locatello, F., Bauer, S., Ke, N. R., Kalchbrenner, N., Goyal, A., & Bengio, Y. (2021). Toward causal representation learning. *Proceedings of the IEEE*, 109(5), 612–634. <https://doi.org/10.1109/JPR OC.2021.3058954>
- Shen, X., Liu, F., Dong, H., Lian, Q., Chen, Z., & Zhang, T. (2022). Weakly supervised disentangled generative causal representation learning. *Journal of Machine Learning Research*, 23, 241–124155.
- Squires, C., Seigal, A., Bhate, S., & Uhler, C. (2023). Linear Causal Disentanglement via Interventions
- Varici, B., Acartürk, E., Shanmugam, K., Kumar, A., & Tajer, A. (2023). Score-based Causal Representation Learning with Interventions. arXiv preprint arXiv:2301.08230
- Varici, B., Acartürk, E., Shanmugam, K., Kumar, A., & Tajer, A. (2025). Score-based causal representation learning: Linear and general transformations. *Journal of Machine Learning Research*, 26(112), 1–90.
- Wendong, L., Kekić, A., Kügelgen, J., Buchholz, S., Besserve, M., Gresele, L., & Schölkopf, B. (2023). Causal component analysis. In: Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023.
- Yang, M., Liu, F., Chen, Z., Shen, X., Hao, J., & Wang, J. Causalvae: Disentangled representation learning via neural structural causal models. In: IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2021, June 19–25, 2021, pp. 9593–9602. Computer Vision Foundation / IEEE, Virtual Event (2021). <https://doi.org/10.1109/CVPR46437.2021.00947>
- Yang, S., Yu, K., Cao, F., Liu, L., Wang, H., & Li, J. (2021). Learning causal representations for robust domain adaptation. *IEEE Transactions on Knowledge and Data Engineering*. <https://doi.org/10.1109/TKDE.2021.3119185>
- Yu, K., Guo, X., Liu, L., Li, J., Wang, H., Ling, Z., & Wu, X. (2020). Causality-based feature selection: Methods and evaluations. *ACM Computing Surveys*. <https://doi.org/10.1145/3409382>
- Yu, Y., Chen, J., Gao, T., & Yu, M. (2019). DAG-GNN: DAG structure learning with graph neural networks. In: *Proceedings of the 36th International Conference on Machine Learning*, ICML 2019, 9–15 June 2019. Proceedings of Machine Learning Research, vol. 97, pp. 7154–7163. PMLR.
- Zhang, J., Greenewald, K.H., Squires, C., Srivastava, A., Shanmugam, K., & Uhler, C. (2023). Identifiability guarantees for causal disentanglement from soft interventions. In: Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023.
- Zheng, X., Aragam, B., Ravikumar, P., & Xing, E. P. (2018). Dags with NO TEARS: continuous optimization for structure learning. *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems NeurIPS* (pp. 9492–9503)

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